# MoCHII Among the Monte Carlo Photoionization Codes: An Algorithmic and Physical Comparison with M$^3$, MAPPINGS V 5.2.1, IONIZE, CMacIonize, Torus, Mocassin, and Colt


**Abstract**

This memo compares MoCHII (MOnte Carlo for H II regions) with the Monte Carlo photoionization codes closest to it in method, and with the deterministic one-dimensional plasma code two of them descend from. The principal pairwise comparison is with M$^3$ (Messenger Monte Carlo MAPPINGS V) and its parent MAPPINGS V 5.2.1, whose local trees are `~/RT_Codes/MAPPINGS/M3-main` and `~/RT_Codes/MAPPINGS/mappings_V-521`. Section 5. adds four further three-dimensional codes: IONIZE, the code of Wood, Mathis & Ercolano (2004) that supplies the WME column of the Lexington benchmark tables; CMacIonize, which its own source describes as the C++ version of that code; Torus, a radiation-hydrodynamics code whose photoionization loop borrows from both IONIZE and Mocassin; and Colt, whose 2025 update aims at the same gas layer MoCHII builds and whose source says several things its paper does not. Mocassin is MoCHII's quantitative validation reference and appears throughout. Section 10.2 compares the one emission process all of them compute, the hydrogen recombination lines, together with Cloudy c25.00, which is the one code in reach that solves the hydrogen atom instead of reading a table. There are four method classes among the seven, and no two codes agree on how the case A/B question should even be posed.

The comparison rests on connected execution paths and quantitative tests rather than on routine names. MoCHII is strongest as a modern three-dimensional transport and synthetic-observation engine: octree AMR, an analytic absorption estimator, a dedicated Cartesian DDA backend, solution-driven ionization-front refinement, dust absorption and scattering, stochastic dust/PAH emission, and observer peel-off imaging; and it is the only code here whose thermal balance and emergent line list come from the same level populations at the same electron density. MAPPINGS V 5.2.1 is strongest as a deterministic one-dimensional general plasma engine: a 30-element option, density-dependent multilevel cooling, finite-source-age photoionization, non-equilibrium cooling, radiation pressure, and steady radiative MHD shocks with an iterated precursor. It also has the most physical hydrogen-line treatment of the six three-dimensional and one-dimensional nebular codes compared here, a case parameter computed from the local Lyman-line escape probability rather than chosen. M$^3$ occupies a distinct middle ground: it grafts three-dimensional Monte Carlo transport onto an older MAPPINGS V 5.1.13-era physics tree, so it combines broad MAPPINGS microphysics with arbitrary Cartesian geometry, but it does not inherit all later 5.2.1 changes or expose all parent modes through its public 3D driver.

The limitations are correspondingly different. MoCHII carries eleven metals to stage VII only, with no coronal stages and no Auger stage coupling; M$^3$ has no AMR, a communication-heavy uniform-grid packet algorithm, and a 3D driver that hardcodes pure case B; MAPPINGS V 5.2.1 is restricted to spherical or plane-parallel 1D structures and uses an outward-only diffuse field; IONIZE is serial, dust-free, and has no convergence test; CMacIonize has no global convergence criterion, no dust in the photoionization path, and a temperature solve that is off by default; and Torus has the strongest transport and parallelism of the family alongside the least complete gas microphysics, missing He II photoheating and emitting no helium line at all. The recommended strategy is to retain the MoCHII transport architecture, use MAPPINGS V 5.2.1 as the current microphysics and 1D shock/time-dependence reference, use M$^3$ to

study how MAPPINGS physics behaves after being mapped into three dimensions, and read TORUS for transport and parallel design rather than for gas physics.

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**Current atomic-physics state (2026-10-11).** MOCHII carries stages I–VII of all eleven metals (77 stages, 66 transitions; `data/atomic/element_*.txt`), which is the stage scope of MOCASSIN v2 (`nstages = 7`). A band whose upper edge lies above the terminal ionization potential of a present element is refused (the lowest is Cl VII, 114.2 eV; Ar VII 124.4, Fe VII 125.0 eV). Metal photoabsorption is Verner's `phfit2` sum: the VFKY96 outer-shell fit below the first inner-shell edge, the sum of the Verner & Yakovlev (1995) subshell fits at and above it, with no option to keep the outer shell alone; the ground term of Ne III takes the Nahar (2019) *R*-matrix cross section below 178.42 eV. The default excitation model (`par%metal_excitation_model = 'chianti'`) is the full CHIANTI 11.0.2 atoms: 70 atoms with 6629 levels and 70 cooling fits, 67 stages from CHIANTI and Cl VII, Ar II, Ca IV from the Stout files of CLOUDY c25.00; C VII, Ar I, Cl I, Ca I, Fe I, Cl V and Ca III have no line cooling. `'mocassin_v2'` (the MOCASSIN v2 atoms, 55 atoms with 571 levels and 49 cooling fits, plus the CHIANTI Fe II forbidden rows) is kept for parity runs; `'extended'` is refused. Every stage with an atom cools and emits. Metal collisional ionization, atomic-hydrogen impact on the C I, C II and O I fine structure, charge-exchange heating and the He I $2^3$S coupling are on by default; the FUV band (from 4.78 eV) is on whenever He I $2^3$S, a metal with a neutral edge below 13.598 eV, or dust is present. The transported diffuse field adds the metal ground free–bound continua (channel 6) and the free–free of every ion inside the band (channel 7). Grain photoelectric heating is Weingartner & Draine (2001) and the neutral gas–grain exchange Hollenbach & McKee (1979), both on with dust. The record is `md/ATOMIC_PHYSICS_RESTORATION_RESULTS_2026-10-11.md`.

*Benchmark numbers.* The converged Lexington runs on this state, in `results/continuous_energy/m13_restoration/` (restarted from the converged run `m11` of 2026-09-27; 29 and 26 iterations to the drift test), give $\langle T[N_pN_e]\rangle$ = 8176 and 6942 K (HII40, HII20; –0.41% and +0.46% from CLOUDY c25.00), $L(\mathrm{H}\beta) = 2.025 \times 10^{37}$ and $4.85 \times 10^{36}$ erg s$^{-1}$. Values from runs that were not repeated — the line ratios of the M$^3$ comparison below among them — are tagged "(m11; not rerun under the 2026-10-11 physics)". The PN150 run stopped unconverged on 2026-10-12 and is to be rerun, and the places that need it say so. No matched full spatial comparison with MOCASSIN v2 has yet been made.

# 1. Purpose, scope, and basis of the comparison

MOCHII and M$^3$ are self-consistent three-dimensional Monte Carlo photoionization codes, whereas MAPPINGS V 5.2.1 is a deterministic one-dimensional plasma and radiative-shock code. MOCHII adds gas physics to the validated MoCafe v2.00 dust radiative-transfer engine. Its founding design treats grid traversal, radiation estimators, dust physics, parallelism, file formats, and synthetic observations as first-class components. M$^3$ adds three-dimensional Monte Carlo radiation transport to MAPPINGS V, whose principal strength is its extensive atomic, ionic, heating, cooling, and emission-line physics [1, 3]. MAPPINGS V 5.2.1 continues that parent line as version 5.2.1, with separate photoionization, shock, equilibrium-cooling, non-equilibrium-cooling, and single-ion diagnostic drivers.

This memo distinguishes three levels of evidence:

1. *Physics present in the parent code.* The MAPPINGS V source tree contains one-dimensional photoionization, shock, non-equilibrium, dust, and plasma utilities.

2. *Physics connected to the three-dimensional driver.* A routine appearing in the source tree is not automatically active in the `montph7` Monte Carlo path.

3. *Physics quantitatively gated in the current code.* This is the strongest evidence: an analytic test, independent solver, database comparison, or published benchmark with a stated numerical tolerance.

This distinction is essential for shocks and time dependence. The public MAPPINGS tree contains `shock2`--`shock5`, `timion`, and `timtqui`; however, the three-dimensional `montph7` family sets `nmod='EQUI'` before the cell thermal solve. Thus the present three-dimensional M$^3$ calculation is an equilibrium photoionization model. The legacy routines remain valuable MAPPINGS capabilities, but they should not be described as coupled three-dimensional M$^3$ physics without additional development and validation.
No direct same-input, same-atomic-data timing or line-flux campaign among the three local executables has yet been performed. Numerical performance comparisons in this memo therefore use algorithm structure and each code's own recorded gates. They are diagnostic rather than a substitute for a controlled cross-code experiment.

## 2. Lineage: why MAPPINGS V 5.2.1 is not M$^3$ 5.2.1

The three code names can easily obscure the most important architectural fact. M$^3$ is not a three-dimensional front end to the newly added MAPPINGS V 5.2.1 directory. Its source identifies itself as MAPPINGS V 5.1.13 with a separate Monte Carlo development history. The two trees still share 78 Fortran filenames, but the shared implementations have diverged substantially; the 5.2.1 versions of core files such as `photo6.f`, `photo7.f`, `shock5.f`, `mapinit.f`, and `cool.f` contain extensive later changes. M$^3$ adds its own `montph7`, Cartesian packet transport, PDF, and three-dimensional thermal-solver files instead [1, 4].
This gives each branch a different meaning:

- MAPPINGS V 5.2.1 is the newer parent-physics branch. Its public `map52` executable offers P6/P7 multizone photoionization, S5 shock plus precursor models, CIE and NEQ cooling curves, PIE curves, single slabs, and multilevel-ion diagnostic modes.

- M$^3$ is the three-dimensional geometry branch. It is the relevant comparison for arbitrary density fields, but its physics inventory must be assessed from the connected `montph7` path, not inferred from the presence of a similarly named MAPPINGS routine.

- MOCHII is an independent three-dimensional transport branch built from MoCafe rather than MAPPINGS. Similar physical terms therefore have independent algorithms, atomic-data preparation, and validation histories.

Consequently, a discrepancy between MOCHII and M$^3$ cannot safely be labeled a discrepancy with MAPPINGS V 5.2.1 5.2.1. For future differential tests, 5.2.1 should be the primary one-zone and 1D microphysics reference; M$^3$ should be treated as the older MAPPINGS-derived 3D realization.

# 3. MAPPINGS V 5.2.1 in its own right

## 3.1 Deterministic 1D integration and diffuse radiation

P6 and P7 integrate successive zones in spherical or plane-parallel geometry. Step sizes respond to optical depth and predicted changes in temperature and ionization. The result has no Monte Carlo shot noise and is therefore efficient for large parameter grids, detailed spectra, and thin cooling zones. Isobaric models can include radiation pressure, $P(r) = P_0 + P_{\rm rad}(r)$, while other choices provide constant density or a prescribed density law. The model may terminate at a distance, an ion or hydrogen column, or an optical/radiation-bounded condition.

The price is geometric closure. The diffuse field is explicitly labeled "OUTWARD ONLY" in `photo6.f` and `photo7.f`. It follows the allowed 1D directions rather than transporting recombination photons over arbitrary solid angle. This is fast and often adequate for a monotonic sphere or slab, but it cannot represent porous escape, oblique shadows, clump illumination, or inward/backward diffuse coupling as a true 3D transport calculation can. The word "adaptive" in the S5 menu refers to an adaptive one-dimensional shock integration, not to a 3D AMR mesh.

## 3.2 Time dependence, cooling flows, and shocks

Unlike the public M$^3$ 3D driver, the P6/P7 path genuinely connects a finite source lifetime to the time-dependent ion solver. It supports an equilibrium solution, a finite-age ionization calculation, and a post-equilibrium source-off evolution. The implementation advances ionic populations with adaptive substeps and couples the elapsed ionization-front time to each new zone. This is a useful one-dimensional ionization-history model, although it is not radiation hydrodynamics and does not evolve an arbitrary multidimensional density field.

The S5 path is an even clearer 5.2.1 advantage. It solves a steady, plane-parallel radiative MHD shock, adapts its step to atomic, cooling, and photon-absorption timescales, and iterates the shock radiation with an automatically calculated photoionized precursor. The source description advertises shock speeds from roughly 10 to $10^4$ km s$^{-1}$. Companion drivers compute CIE cooling, non-equilibrium cooling, and PIE curves. For one-dimensional shock spectra and cooling flows, neither current 3D code offers an equivalent connected and mature path.

## 3.3 Atomic, electron, and dust physics

The standard preference file activates 16 elements, while `mapFull.prefs` activates all 30 elements from H through Zn; static dimensions permit 31 ion stages and 10,240 photon-energy entries. The thermal loop directly calls fine-structure, intercombination, resonance, multilevel, iron, recombination, free-free, collisional-ionization, charge-exchange, grain/PAH, and cosmic-ray terms. Optional $\kappa$ electron distributions modify excitation, ionization, and recombination rates. These are major strengths for broad plasma diagnostics and for gas near or above coolant critical densities.

The 1D dust path is materially more complete than the public M$^3$ 3D dust branch. In addition to grain charge, depletion, photoelectric heating, gas-grain collisions, PAHs, and destruction prescriptions, P6/P7 call `dusttemp` zone by zone. That routine implements a Draine–Li stochastic temperature distribution and returns an infrared photon field. Thus MAPPINGS V 5.2.1 has a coupled 1D gas–dust–IR calculation, whereas MOCHII has the stronger spatially resolved 3D dust transport and imaging chain.

Several cautions remain. The atomic tables are broad but globally coupled and are harder to provenance or replace ion by ion than the MoCHII registry. A Compton routine exists but no call from the examined main cooling path was found, and the microturbulent heating block in `cool.f` is commented out; neither should be counted as active 5.2.1 physics without a dedicated execution test. Resonance-line attenuation uses escape/transfer approximations rather than a multidimensional, frequency-redistributing Monte Carlo line solver.

### 3.4 Software and workflow limitations

MAPPINGS V 5.2.1 preserves a productive but legacy Fortran architecture: common blocks, compile-time maxima, runtime table loading, interactive menus, and many ASCII output products. P7 exposes experimental routines and an API, but the normal workflow remains less scriptable and less modular than MoCHII's namelist, HDF5/FITS, module, and Python-reader design. The documentation overview describes the manual as work in progress, so reliable use still requires source reading. The benefit of this conservatism is continuity with decades of MAPPINGS models; the cost is a higher barrier to automated regression testing, embedding, and isolated replacement of atomic data.

## 4. Executive comparison

Table 1: High-level comparison of the current three-dimensional codes.

| Topic | MoCHII | M$^3$ |
|---|---|---|
| Grid | Octree AMR or a memory-minimal uniform Cartesian backend; optional solution-driven I-front refinement. | Uniform Cartesian grid divided into equal MPI blocks; no AMR. |
| Absorption transport | Analytic leaf-by-leaf attenuation estimator when the band has no scattering; only direction and frequency are sampled. | Lucy-style random optical depth, discrete absorption, and immediate local re-emission. |
| Diffuse field | Explicit H/He recombination channels, the metal ground free–bound continua and the free–free of every ion inside the band, rebuilt from the current state and transported (the default since 2026-09-26; the metal and free–free channels since 2026-10-11); on-the-spot remains available. No metal recombination lines. | Local continuum and line emissivity PDF sampled immediately after gas absorption; isotropic re-emission. |

*Table 1 continued.*

| Topic | MoCHII | M$^3$ |
|---|---|---|
| Frequency treatment | Continuous photon energies drawn from the source and diffuse spectra, cross sections evaluated at each packet's energy; the FUV extension has its own packet budget, and the threshold-aligned bins are a diagnostic $J_\nu$ tally only. | Up to 10,240 bins; a fixed number of packets is launched at every frequency and packet energy follows the source spectrum. |
| Parallel model | Packet decomposition and shared-memory grid arrays; one full grid copy per node. | Spatial domain decomposition; packets crossing blocks are synchronized through MPI. |
| Gas microphysics | Modern, provenance-bearing data for stages I–VII of eleven metals; full CHIANTI 11 atoms; line cooling in the thermal loop suppressed to the local $n_\mathrm{e}$ through $\Lambda(T, n_\mathrm{e})$ tables built from its own $n$-level solve. | Broad MAPPINGS ion network and multilevel line cooling inside the thermal loop; older but extensively exercised atomic-data compilation. |
| Dust | EUV/FUV absorption and HG scattering, live dust/PAH survival, equilibrium temperature, SEDust stochastic emission, transport of the reradiated infrared back onto the grains, and IR maps — all from one grain model, whose own size-integrated extinction table feeds the transport, so extinction and emission refer to the same grains (gated); Weingartner–Draine grain charging and photoelectric heating and the neutral gas–grain exchange in the leaf's own field. | Rich grain charge, depletion, photoelectric, and PAH microphysics; the public 3D dust transport branch treats dust extinction/absorption but does not expose an equivalent scattering plus stochastic-IR imaging chain. |
| Synthetic observations | Leaf emissivities, flux-conserving line and field maps, dust-band maps, multi-observer peel-off images, and spectral cubes. | Extensive line spectrum and cell-resolved emissivities; maps can be made by projection, but no equivalent general observer peel-off layer is evident in the public path. |

*Table 1 continued.*

| Topic | MoCHII | $M^3$ |
|---|---|---|
| Best current use | Dusty, clumpy, spatially resolved H II regions and atomic PDR skins on simulation-derived density fields. | Broad nebular line diagnostics, denser gas, harder PN-like spectra, and problems that benefit from the full MAPPINGS ion and cooling inventory. |

The concise verdict is:

> MoCHII is presently the stronger three-dimensional transport, dust, AMR, and synthetic-observation framework. $M^3$ is presently the stronger broad-spectrum nebular microphysics framework, especially when many ion stages, or level populations solved inside the thermal iteration for every ion, determine the solution.

Table 2: What MAPPINGS V 5.2.1 adds to the original two-code comparison.

| Topic | MAPPINGS V 5.2.1 strength | Limitation relative to the 3D codes |
|---|---|---|
| Photoionization | Noise-free adaptive 1D integration; equilibrium, finite-age, and source-off evolution; radiation-pressure options. | Only spherical or plane-parallel structure; outward-only diffuse field. |
| Shocks and cooling flows | Connected steady radiative MHD shock plus iterated precursor, CIE cooling, NEQ cooling, and PIE modes. | Steady or prescribed 1D histories, not multidimensional radiation hydrodynamics. |
| Microphysics | Full 30-element preference set, many ion stages, density-dependent line cooling, $\kappa$ electrons, cosmic rays, and broad spectral output. | Global tables are harder to audit and update ion by ion; not every dormant routine is connected to the main cooling path. |
| Dust | Grain/PAH charge and destruction, gas–grain exchange, stochastic temperature distribution, and a coupled 1D IR field. | No arbitrary 3D dust geometry, shadowing, AMR, or observer image construction. |
| Numerics and workflow | Fast deterministic parameter grids with decades of model heritage. | Legacy interactive common-block architecture, compile-time maxima, fragmented ASCII products, and incomplete manuals. |

*Table 2 continued.*

| Topic | MAPPINGS V 5.2.1 strength | Limitation relative to the 3D codes |
|---|---|---|
| Best current use | Integrated spectra of 1D H II regions, PNe, AGN slabs, radiative shocks, and equilibrium/non-equilibrium cooling sequences. | Not a replacement for 3D modeling when geometry is part of the physical question. |

# 5. The Monte Carlo photoionization codes closest in method

M$^3$ is not the only three-dimensional Monte Carlo photoionization code this comparison has to account for. Four others are close enough in method that MOCHII's choices are best read against them: IONIZE, the code of Wood, Mathis & Ercolano (2004) that supplies the WME column of the Lexington benchmark tables; CMACIONIZE, which its own source calls "the C++ version of Kenny Wood's photoionization code" (`src/CMacIonize.cpp`:392–393); and TORUS, a radiation-hydrodynamics code whose photoionization loop borrows from both IONIZE and MOCASSIN; and COLT, whose 2025 update aims at the same gas layer MOCHII builds. MOCASSIN itself is MOCHII's quantitative validation reference and appears throughout this memo rather than as a separate subsection.

The evidence standard of Section 1. applies unchanged: every statement below was read in the source, and where a capability exists in a tree but is not reached from the photoionization driver it is labeled as such. The *cooling* inventories of these same four codes were audited separately and are not repeated here; that audit, with its file and line references and the defects it found on each side, is `docs/MoCHII_cooling_analysis.pdf`. This section covers the other axes.

## 5.1 IONIZE: the reference implementation of the method

IONIZE is the ancestor of this family and the most instructive comparison precisely because it is small: 33 fixed-form Fortran files, one executable, no libraries.

Packets carry photon number rather than energy, and the band is 13.6–54.4 eV only (`planckset.f`:19–21). The estimator is Lucy's path-length form, but *frequency-unresolved*: as a packet crosses a cell, the code accumulates the rate integrals themselves — one accumulator for each species it knows about, `jmeanH`, `jmeanHe`, `jmeanN(...,1:3)` and so on (`tauint.f`:184–268) — and never stores a mean intensity. There is no $J_\nu(\nu,\text{cell})$ array anywhere in the code. The consequence is structural rather than numerical: a rate integral for any species not in that hardcoded list cannot be recovered after the run, so adding an ion is a code operation. MOCHII also accumulates the rate integrals directly — each path segment is scored against the cross sections at the packet's own photon energy (`src/ion_score_mod.f90`, `score_ion_path`) — but it loops over whatever stages the registry holds, so adding an ion stays a data operation; its band tally `jt_ion(nnu,nleaf)` is a diagnostic spectrum, kept only when $J_\nu$ is written (`par%jnu_output`) or the SEDust emission reads it, and no rate is taken from it.

The opacity is hydrogen and helium only (`tauint.f`:159–161); the metals are tallied but never attenuate. The diffuse field, by contrast, is fully explicit and physically consistent: the balance uses the Verner & Ferland case-A total

recombination coefficient (`recomb.f`:13–15, $4.19 \times 10^{-13}$ cm$^3$ s$^{-1}$ at $10^4$ K), and after each absorption the packet is re-emitted as a ground-state continuum photon with probability $\alpha_1/\alpha_A$, sampled from a Saha–Milne emissivity distribution, with four helium channels and He I Lyman $\alpha$ handled on the spot (`probset.f`:28–60, `mcionize.f`:278–339).

The grid is a uniform Cartesian mesh with $65^3$ compiled in (`grid.txt`:3), and the density is a hardcoded analytic function of position evaluated on cell centers (`density.f`:15–20); there is no grid-file reader, so a simulation-derived density field cannot be read at all. The code is *serial*: no MPI and no OpenMP appear in any source file. Seven elements and 21 stages are followed, without He$^{++}$, and with C$^0$ and S$^0$ assumed absent; $n_e$ comes from ionized hydrogen and helium only, which the source states outright (`tbal1.f`:39–40).

Two properties of its iteration are worth recording because MOCHII spent real effort on both. There is *no convergence test* — the manual prescribes running many iterations and comparing successive ones by eye — and the ionization is iterated alone for the first four iterations, the thermal balance starting only at the fifth (`itbal.f`:27–41). The temperature solve is a logarithmic secant on the gain/loss ratio (`tbal1.f`:78–107); below 4000 K the cell is declared neutral and pinned at 500 K.

There is no dust. The only grain-related term is a scalar gas-heating factor $3 \times 10^{-25}\, n n_e$ scaled by an input parameter that the shipped deck sets to zero (`ioneng.f`:56–68). Outputs are a direction-binned escaping spectrum on a $20 \times 20$ grid of $(\theta, \phi)$ and an emissivity cube; there is no imaging, and the manual's recipe for a map is to sum the cube along an axis. The sibling directory `diffuse/` in the same distribution is a different code, a dust-scattering polarization Monte Carlo with Stokes peel-off imaging and no gas physics; it shares only utility routines and should not be read as IONIZE's imaging stage.

## 5.2 CMACIONIZE: the same method, modernized

CMACIONIZE inherits IONIZE's physics and rewrites its software. The transport is a random optical depth with the Lucy path-length estimator (`src/IonizationPhotonShootJob.hpp`:135, `src/DensityGrid.hpp`:150–197), the diffuse re-emission is the Wood–Mathis–Ercolano scheme reimplemented term by term (`src/PhysicalDiffuseReemissionHandler.cpp`:52–203), and the hydrogen line coefficients are numerically identical to IONIZE's (Section 10.2). What changed is everything around the physics.

Frequencies are continuous: a packet's cross sections are evaluated once at emission (`src/PhotonSource.cpp`:230–244) and there is no band grid in transport at all. This removes the band-discretization question — which biased MOCHII's helium and metal ionization while it transported band-averaged energies, and which MOCHII has since removed the same way (Section 6.3) — and replaces it with sampling noise in the rate integrals. Three grid types are offered — uniform Cartesian (the default), an octree AMR with Morton-style keys, and a Voronoi mesh that can move with the fluid — plus a fourth structure, a subgrid decomposition used by the task-based mode. Refinement can be static or solution-driven, on either cell opacity or O I content (`src/OpacityAMRRefinementScheme.hpp`:90–99, `src/OIAMRRefinementScheme.hpp`:87–94). Two gaps are visible in the source: `AMRRefinementScheme::coarsen` is declared and has no caller anywhere, so the mesh can only ever get finer, and the MPI branch of the refinement carries the comment `/// WE HAVE TO DO SPECIAL THINGS FOR MPI HERE!` (`src/AMRDensityGrid.hpp`:337).

Parallelism is OpenMP through a job market plus MPI as a photon-count decomposition over a *fully replicated* grid, with the accumulators reduced across ranks and the cell solve split by contiguous cell block

(`src/IonizationSimulation.cpp`:395–397, 458–617). That is the same division of labor as MOCHII's, which also replicates the grid and distributes packets; TORUS is the one code here that decomposes the grid itself. MPI is refused at startup for the dust, hydrodynamic, emissivity and task-based modes.

Three properties bear directly on the comparisons this memo makes elsewhere. The metal line emissivities are solved from level populations at the local electron density (`src/LineCoolingData.cpp`:1785–1798), which is the axis MOCHII closed with `par%cooling_model = 'local_ne'`. The temperature solve is *off by default* and, when on, is capped at 30 000 K for a stated reason — "helium charge transfer rates are only valid until 30,000 K" (`src/TemperatureCalculator.cpp`:831–833). And there is *no global convergence criterion of any kind*: the iteration is `while (loop < _number_of_iterations)` (`src/IonizationSimulation.cpp`:359), with a plain input count, no cell-maximum test, no integrated measure and no early exit. Metals carry neither opacity nor free electrons, there is no collisional ionization anywhere in the tree, and $He^{++}$ does not exist as a species.

Dust is absent from the photoionization path — a case-insensitive search of the ionization, temperature and state-calculator sources returns nothing. It exists instead as a separate hardwired mode whose own documentation calls it a first-year lab project (`src/CMacIonize.cpp`:50–53), with gray constants in each band, a forced first interaction, single-observer peel-off and Stokes imaging. There is no dust temperature and no infrared emission anywhere in the tree.

Synthetic observations are the weakest axis. The ionization mode offers point "trackers" that record the absorption or the spectrum inside a named cell, and grid dumps whose field list contains no emission-line entries. A line-of-sight emission integrator, `get_total_emission`, is implemented for all three grid types and *has no caller*; the emissivities it would integrate are themselves produced only by a separate post-processing mode, which is Cartesian-only and rejects MPI. Two statements in the tree's own documentation are stale and should not be quoted from it: the claim that the code is parallelized with OpenMP only (`src/CMacIonize.cpp`:416–420), contradicted by the working MPI code, and a comment implying a Python library replaced the in-run emissivity calculation when the compiled mode is the fuller route.

## 5.3 TORUS: photoionization inside a radiation-hydrodynamics code

TORUS is the most capable of the three in transport and the least specialized to photoionization, which shows in both directions.

The estimator is again Lucy's, with the path length weighted by $\ell w/(h\nu)$ and accumulated for each ion (`photoionAMR_mod.F90`:6679–6692); a modified random walk handles the optically thick regime. Dust scattering uses real Mie phase matrices rather than a Henyey–Greenstein approximation (`:3283`). The diffuse field is transported by default, and the switch that turns it off removes only the ionizing continuum from the redraw spectrum (`:3347--3349`), leaving the sub-Lyman lines and the dust continuum in place — so "on the spot" means something narrower here than in MOCHII. Independently of that, a `caseB` flag replaces the hydrogen recombination coefficient with the *constant* $2.7 \times 10^{-13}$ cm$^3$ s$^{-1}$, with no temperature dependence (`photoion_utils_mod.F90`:305–310); setup only warns if it is combined with a transported diffuse field. This is exactly the conflation MOCHII separated into `par%recombination_continuum` and `par%lyman_lines`, left here as a warning rather than as two parameters.

The frequency grid is 1000 logarithmic bins from 1 mm to about 124 eV with four extra points pinned at $\pm 1\%$ around the helium edges (`:2124--2151`) — threshold *bracketing*, the same concern as MOCHII's threshold alignment, addressed

by refining around the edge rather than by placing a bin boundary on it.

The grid is an AMR tree that can be one-, two- or three-dimensional (binary, quad, octree), with a cylindrical option and a two-dimensional Cartesian slab whose missing third dimension is compensated by re-weighting the emission direction. Solution-driven refinement is live and physically the same criterion MOCHII uses: a relative neutral-hydrogen-fraction jump across each face against a tolerance (`hydrodynamics_mod.F90`:14582–14618). One switch is dead and should not be cited as a capability: `doPhotoRefine` is read from the input file and its only use site is `if (doPhotoRefine .and. 0 == 1)` (`:4428`).

Parallelism is the most developed of the four: MPI decomposes the grid itself, with packets shipped between ranks in batched stacks, plus extra ranks assigned as duplicates of busy domains apportioned by measured packet crossings (`loadbalance_mod.F90`:16–219), and OpenMP threads the equilibrium solve over cells (`:3855--3864`). MOCHII distributes only transport and keeps one grid copy per node; TORUS distributes the state.

Convergence is a nested pair: an inner alternation of ionization and thermal solves until both change by less than 1% in every subcell, and an outer test on the *cell-maximum* fractional temperature change below 0.05 (`:4324--4339`). There is no volume-integrated measure. What TORUS does instead, when the outer test stalls, is diagnose the cause: cells the Monte Carlo has undersampled are counted, and finding any of them doubles the packet count and continues (`:4344--4347`). That is an answer to the same noise-floor problem MOCHII solved with `par%conv_crit = 'vol'`, from the other end — raise the photon budget until the cell-maximum test becomes meaningful, rather than change the measure.

The heating inventory is exactly three terms: H I photoionization, He I photoionization, and dust (`:7810--7823`). He II photoheating is absent although $He^{++}$ is tracked, and there is no metal photoheating and no grain photoelectric heating in this balance. Recombination data are Verner & Ferland for H and He and Pequignot et al. (1991) plus Nussbaumer & Storey (1983) for the metals — *the* MOCASSIN *set*, so the in-code swap experiment reported in Section 8.2 transfers to TORUS unchanged. Its $H^+$ and $He^{++}$ coefficients (Verner & Ferland, `photoion_utils_mod.F90`:304–335) sit 0.72% and 0.27% above MOCHII's default $\alpha_A$ at $10^4$ K, which since 2026-09-26 is the Storey & Hummer (1991) cascade $\alpha_B$ plus the Milne $\alpha_1$ (they agreed to four digits with the Badnell total MOCHII used before); $He^+ + e$ differs most, TORUS sitting 4.7% *above* MOCHII's Badnell total, the opposite side from MOCASSIN's low value. One provenance curiosity: the free-bound $\gamma$ tables TORUS reads are byte-identical to MOCASSIN's and MOCHII's copies, but its own comment attributes them to Ferland (1980) where the other two carry the Ercolano & Storey compact format. The numbers are not in question, only the citation.

Dust is coupled to the gas in both directions, which no other code in this comparison does: grain emission enters the gas cooling side and grain heating the gas heating side of the same balance, or, decoupled, a gas–grain collisional exchange term carries the energy between them (`:6604--6628`, `:6253--6271`). The infrared emission is a single equilibrium temperature per cell; there is no stochastic single-photon heating and no PAH channel on this path, so MOCHII's SEDust coupling and its $x_{\rm HII}$-weighted PAH share have no counterpart.

Imaging is genuine peel-off with the full Stokes vector and $e^{-\tau}$ to the boundary, cloned at emission and after every scattering (`:9109--9113`, `image_mod.F90`:187–188), for one of four image types (free-free, forbidden line, recombination line, dust). It is monochromatic: the photoionization image path carries a fixed wavelength and applies no Doppler shift. TORUS does have a velocity-resolved image adder, but it belongs to the atomic and molecular line transfer modes and is

not wired to the photoionization loop — present in the tree, not connected. There is no emergent SED and no datacube from this loop.

## 5.4 What the four have in common, and where MoCHII sits

Table 3: The Monte Carlo photoionization codes closest to MoCHII in method, on the axes not covered by the cooling audit. Entries are what the photoionization path does, not what exists in the tree.

| Topic | IONIZE | CMACIONIZE | TORUS | MOCHII |
|---|---|---|---|---|
| Estimator | Lucy path length, rate integrals accumulated directly; no $J_\nu$ stored | Lucy path length, continuous frequency | Lucy path length, 1000-bin grid | Analytic edge walk without scattering, continuous photon energy, rates scored at the packet energy; $J_\nu$ tally on request |
| Grid | Uniform Cartesian, size compiled in; analytic density only | Cartesian / octree AMR / Voronoi / subgrid | AMR in 1D, 2D or 3D, cylindrical option | Octree AMR or raster Cartesian with a DDA walk |
| Solution-driven refinement | none | opacity or O I criterion; no coarsening | $x_{\rm HI}$ gradient across faces | $x_{\rm HI}$ front flags plus face-neighbor jump |
| Parallel model | serial | OpenMP + MPI over packets, grid replicated | MPI over the grid with load-balancing ranks, OpenMP over cells | MPI over packets, shared-memory grid per node |
| Diffuse field | explicit, case A, four He channels | the same scheme, opt-in, default off | transported by default; `caseB` is a constant coefficient | explicit H/He ground continua, the He I and He III excited cascades, metal ground continua and free–free, transported by default; on-the-spot available |
| Global convergence | none (compare iterations by eye) | none (fixed iteration count) | cell-maximum $\Delta T_{\rm e}$, packet doubling on undersampling | cell-maximum or volume-integrated; the volume test extrapolates the decay (rate fitted over 8 iterations) or requires no resolvable signed drift |

*Table 3 continued.*

| Topic | IONIZE | CMACIONIZE | TORUS | MOCHII |
|---|---|---|---|---|
| Metals | tallied, no opacity, no electrons | tallied, no opacity, no electrons; no collisional ionization | 22 ions, no opacity, electrons counted, charge exchange data for five ions only, no collisional ionization | registry with opacity, electrons and heating (on by default), two collisional-ionization models |
| Dust | none | none on this path | absorption, Mie scattering, equilibrium $T_{\rm dust}$, gas–grain exchange | absorption, scattering, live survival, $T_{\rm dust}$, stochastic PAH emission, grain photoelectric heating, gas–grain exchange |
| Images | escaping spectrum by direction bin; emissivity cube | trackers only; the line-of-sight integrator is never called | Stokes peel-off, monochromatic, four image types | multi-observer peel-off, direct/scattered split, bin-resolved cubes, flux-conserving line maps |

Read across that table, the family divides on one line. IONIZE and CMACIONIZE are photoionization codes in which the radiation field is a means to the ionization state, and their synthetic-observation layer is correspondingly thin. TORUS is a transport code with photoionization added, which is why its imaging and parallelism are the strongest here and its gas microphysics the most incomplete — missing He II photoheating and, as Section 10.2 shows, missing helium lines entirely. MOCHII sits with TORUS on transport and observation and with MOCASSIN on gas microphysics, and it is the only one of the five whose thermal balance and emergent line list are computed from the same level populations at the same electron density. COLT is described as acquiring that property in 2025, but the level-population solver is not in its public tree, so the claim stands unqualified for now; Section 5.5 reads that code against the same axes. How each of them actually solves the ionization state, and where the metals enter it, is Section 8.3.

## 5.5 COLT: the fifth code, and what its source says that its paper does not

COLT (Cosmic Lyman-$\alpha$ Transfer) is read here in its own tree, `~/RT_Codes/colt` at `d787fb8` (2026-08-21), on the same standard as the four above. The published description is McClymont, Smith & Tacchella (2025) [6], a preprint and not yet a published paper; where the source and the paper disagree, this section follows the source and says so. COLT belongs in this comparison because its 2025 update is aimed at exactly MOCHII's layer: an ionization solver over ten elements, a nebular continuum from free–free, free–bound and two-photon emission, and metal line emission from level populations.

**What is in the tree and what is not.** The ionization solver, the rate compilation, the Verner cross sections, the free-free Gaunt factors, the nebular continuum and the diffuse-source construction are all present and are read below. The *thermal* equilibrium solver of the paper's Sections 2.5–2.6 is not: `grep` for `bisect` over the tree returns nothing, there is no

cooling function, no Compton term, and no CHIANTI data anywhere. What `equilibrium_update` does with the temperature is reconstruct it from the internal energy (`ionization/ionization.cc`:1946–1960), with multiplicative limiters `T_inc_factor` / `T_dec_factor` on how far it may move in one pass. The level-population solver is likewise absent. By the hierarchy of Section 1. the thermal solve, the CHIANTI level populations and the Courant-limited cooling are level-1 evidence for COLT — described, not connected in this checkout — and the rest of this subsection is level 2 or better.

**The ionization solve is the same ratio chain, and hydrogen carries no charge exchange.** This is the sixth code read for Section 8.3 and it does not change that section's finding. Hydrogen is `cHI = (bHI + gHI/ne)/aHII`, `xHI = 1/(1 + cHI)` (`:1901--1902`) — collisional plus photoionization over recombination, with no charge transfer term, exactly as in MOCASSIN, CMACIONIZE, TORUS and IONIZE. Helium carries one against H I in its recombination denominator (`:1913--1914`), one-way. The metals are a nearest-neighbor ratio chain written as a recurrence, `c_ions[ai] = (beta + chi_HII HIIe + chi_HeII HeIIe + gamma/ne) / (alpha + xi_HI HIe + xi_HeI HeIe)` then `denom = 1 + c_ions[ai] denom` (`:1926--1928`), one element at a time inside a fixed-point loop on $n_e$ that is damped by `xe = 0.5 (xe + xe_prev)` (`:1942`) and exits at $|\Delta x_e| < 10^{-9}$ or errors out after 100 passes. So COLT sits with MOCHII's `'metal_only'` mode, not with its default. On one axis it is ahead of MOCHII: its charge exchange runs against four partners, H I and He I for recombination and H II and He II for ionization (Kingdon & Ferland 1996), where MOCHII carries the hydrogen pair only for the metals (its He–H exchange, added 2026-09-24, couples helium to hydrogen, not the metals to helium).

**Metal opacity: COLT has it.** `k_gas_abs` sums $n_{\rm ion}\sigma_{\rm ion}$ over every active ion of every active atom (`ionization/ray_trace.cc`:1492–1505), so the ionizing field is attenuated by the metals and not merely tallied against them. That places COLT with MOCHII and apart from CMACIONIZE, TORUS and IONIZE on the axis of Section 8.3. Absorption is continuous (`weight *= exp(-dtau_abs)`, `:1456`) with a path-length rate estimator; dust scattering is by sampled optical depth.

**The diffuse field: the paper understates what the code does.** The paper says of cell-based nebular continuum emission that "our current implementation uses Case B rates which assume on-the-spot absorption of the ionizing component, and so it is not appropriate to incorporate this emission, although we plan to relax this assumption in the future." The ionization module nevertheless builds and launches it. Three flags, `free_free`, `free_bound` and `two_photon`, all `false` by default (`ionization/vars.cc`:16–18), turn on cell-based sources whose frequency is drawn from a bin cumulative distribution assembled from free–free for $Z = 1$ (H II and He II) and $Z = 2$ (He III), free–bound for H II→H I, He II→He I and He III→He II, and two-photon for H I and He II (`ionization/ray_trace.cc`:479–551). The free-bound tables are Ercolano & Storey (2006), the same source as MOCHII's, and they stop below 1 Ryd, so they carry the $n \geq 2$ continuum and do not duplicate the ground-state recombination that the hard-coded $\alpha_B$ assumes on the spot.

Read against MOCHII, that splits two ways. MOCHII is ahead on how much of the recombination cascade is resolved: it samples the three ground continua from the Milne free-bound spectrum with the transport cross sections, splits the He I cascade by temperature-dependent branch fractions, and carries the He III case-B cascade as a fifth channel worth 44% of the diffuse-field luminosity on a $10^5$ K blackbody, none of which has a counterpart here. COLT was ahead on one channel until 2026-10-11: free–free above 13.598 eV was a diffuse ionizing source in COLT and not in MOCHII. MOCHII now transports it as diffuse channel 7 ($H^+$, $He^+$, $He^{2+}$ and the metal ions), from the same coefficient as its

written continuum, the arrangement COLT uses; it also carries the metal ground free–bound continua as channel 6, which COLT does not. Measured on the HII40 geometry (2 iterations, not converged) channel 7 is $4.5\times10^{-10}$ of the diffuse luminosity, and $2.1\times10^{-8}$ on a $10^5$ K blackbody.

**H I and He I recombination.** $\alpha_B(H\,\text{II})$ is the Hui & Gnedin (1997) fit (`rates.h`:101), giving $2.5918\times10^{-13}$ at $10^4$ K, on the canonical value to 0.07%. $\alpha_B(He\,\text{I})$ is the companion power law $1.26\times10^{-14}\lambda^{0.75}$ (`:118`) plus a dielectronic term that is negligible below $10^5$ K; together they give $2.616\times10^{-13}$ at $10^4$ K, which is **0.947** times MOCHII's case-B $\alpha_B(He\,\text{I}) = 2.763\times10^{-13}$ (the Badnell total $4.3719\times10^{-13}$ less the Milne $\alpha_1$), 0.938 at 8000 K and 1.001 at $2\times10^4$ K. An earlier version of this paragraph set the same $2.616\times10^{-13}$ against the case-*A* totals — MOCHII's $4.3719\times10^{-13}$ and the $4.3583\times10^{-13}$ CLOUDY derives level by level from the Milne relation — and called it 0.60; that compared a case-B coefficient with case-A ones, and the factor does not exist. This memo carries the warning that MOCASSIN's $\alpha_A(He\,\text{I})$ is 11% high because it is Verner & Ferland (1996) applied with no special case for helium; COLT's case-B helium coefficient departs the other way, by 5–6% at nebular temperatures, half that size, which is what a single power law fitted for intergalactic reionization bookkeeping leaves. It is worth subtracting before COLT helium fractions are compared with an H II region model, but it is not a large factor. Metal recombination is by contrast current: radiative plus dielectronic from Badnell, stated at `ionization/ionization.cc`:1962 and evaluated ion by ion in `rates.h`.

**Two-photon: COLT keeps the collision partners apart.** The suppression is written out in full, `alpha * A_2q / (A_2q + q_e * ne + q_p * nHII)` (`nebcont.cc`:364), with $q_e$ and $q_p$ each built from the Pengelly & Seaton (1964) expression — reduced masses, $D_{nl} = Z^{-2}\,6n^2(n^2-l^2-l-1)$, and the smaller of the two impact-parameter cutoffs (`:340--363`). $\alpha_{\text{eff}}(2s)$ comes from a two-dimensional Hummer & Storey (1987) table in $(\log T, \log n_e)$; MOCHII takes its own $\alpha_B$ times the $2s$ share of a zero-density $nl$-resolved cascade built on Storey & Hummer's (1991) hydrogenic routines, in case A or B as its recombination lines are, for both H I and He II (until 2026-09-26 the case-B Pengelly 1964 power law for H I and four SH95 anchors for He II). Two differences are worth carrying away.

*The factor near six between the two is a* COLT *defect.* As coded, the COLT expression gives $q_e+q_p = 3.24\times10^{-3}\,\text{cm}^3\,\text{s}^{-1}$ at $10^4$ K and $n_e = 100$, falling to $2.80\times10^{-3}$ at $10^4\,\text{cm}^{-3}$, against the single $5.31\times10^{-4}$ MOCHII takes from Osterbrock's table by way of MOCASSIN. The Pengelly & Seaton cutoff $R_1 = 1.12\hbar v/\Delta E$ needs $\Delta E$, the $2s$–$2p$ separation ($0.0353\,\text{cm}^{-1}$ to $2p_{1/2}$, 0.330 to $2p_{3/2}$); COLT passes `nu_12_HI = c * 82258.9544` (`nebcont.cc`:295), the Ly$\alpha$ frequency. The logarithm of that cutoff goes negative, `minpos` (`:301`) discards it, and only the Debye cutoff is left — hence the weak density dependence. With the physical separations, weighted $1/3$ and $2/3$ by the $2p_{1/2}$ and $2p_{3/2}$ statistical weights, the same expression gives $5.79\times10^{-4}$, independent of $n_e$ over $10^2$–$10^4$; EXHALE's separate Pengelly & Seaton evaluation gives $5.37\times10^{-4}$. The surviving fraction of the H I two-photon continuum at $10^4$ K:

| $n_e$ [cm$^{-3}$] | MOCHII ($n_p = n_e$) | COLT, physical $\Delta E$ | COLT as coded |
|---|---|---|---|
| 100 | 0.994 | 0.993 | 0.962 |
| 1000 | 0.939 | 0.934 | 0.731 |
| 3000 | 0.838 | 0.826 | 0.485 |
| 10000 | 0.608 | 0.587 | 0.227 |

So at the $3000\,\text{cm}^{-3}$ of PN150 the corrected expression and MOCHII agree to 1.5% in the continuum, and COLT as coded removes three times too much of it. Since 2026-09-25 MOCHII also keeps the partners apart

(`par%twophoton_quenching = 'colliders'`): electrons, protons, $He^+$ and $He^{2+}$ each with its own Pengelly & Seaton rate, normalized to Osterbrock & Ferland Table 4.10, with $A_{2s} = 8.2246\,s^{-1}$ (Drake 1986). With protons the only ions, as in the table, that default and the earlier `'lumped_ne'` give the same $q = 5.31\times10^{-4}$ at $10^4$ K, so the MOCHII column holds for both; they part where $He^+$ or $He^{2+}$ collides ($He^+$ at 1.07, $He^{2+}$ at 3.21 times the proton rate) or where metals supply electrons without protons.

*And one slip, stated as read.* `HeII_2p_rate` (`nebcont.cc`:373–375) instantiates the template with the atomic number argument 1 while its own trailing comment reads `// HeII (Z=2)`, so the He II branch takes $A_{2q} = 8.2249\,s^{-1}$ instead of 526.7 and $D_{nl} = 72$ instead of 18. The companion `HeII_2p_freq` (`:401--403`) does pass 2, so the spectral shape is right and only the collisional denominator is affected. The critical density is then 256× too low: at $n_e = 10^4\,cm^{-3}$ the He II two-photon continuum is suppressed to 0.227 where the He II constants give 0.983, and at $10^5$ to 0.031 against 0.863. At $100\,cm^{-3}$ it is 3.8% and does not matter. COLT has not been built or run here, so this is a reading of the source and nothing more; it is recorded because MOCHII transports the same channel through `par%heii_diffuse` and any future comparison of the two would run straight into it. COLT omits He I two-photon emission altogether, commented out at `:404--409` with the note "No rates for HeI (data are poor) but expected to be small"; MOCHII transports and writes it from the Drake, Victor & Dalgarno (1969) distribution as CHIANTI tabulates it. Finally, COLT normalizes the Nussbaumer & Schmutz (1984) shape by the separately quoted $A_{2q}$, `h * x * A_nu / A_2q` (`:393`) — the arrangement MOCHII moved off, because dividing a tabulated shape by a constant it does not integrate to reproduces the pair count only to –0.014% for H I, +0.05% for He II and +0.28% for He I.

**Convergence.** COLT iterates transport against abundances until the relative change in the *total* number of H I recombinations falls below `max_error`, default 0.01 (`ionization/ionization.cc`:2198, `Ionization.h`:27; the paper quotes 0.001 as an example), with an optional pre-conditioning stage that starts at a looser target and tightens geometrically. It is a global integral criterion of the same class as `par%conv_crit='vol'`, applied to one scalar rather than to the ionization state and the temperature.

**Ion inventory.** Ten elements, H He C N O Ne Mg Si S Fe, with the enumeration reaching C VI, N VII, O VIII, Ne VIII, Mg X, Si XII, S XIV and Fe XVI (`source/proto.h`:331–341), against MOCHII's eleven elements at stages I–VII (since 2026-10-11, stage HS2 of Section 16.).

# 6. Radiative-transfer algorithms

## 6.1 Radiation-field estimators

Both three-dimensional codes estimate the cell radiation field from packet trajectories, but their estimators differ fundamentally. M$^3$ follows the Lucy path length estimator. At frequency $\nu$, a packet with energy $\epsilon_\nu$ crossing a distance $l$ in cell volume $V$ contributes schematically

$$J_\nu \;=\; \frac{1}{4\pi\,\Delta t\,V}\sum_{\text{paths}} \epsilon_\nu l. \tag{1}$$

The distance to the next absorption is sampled from

$$\tau_p = -\ln p, \tag{2}$$

with $p$ uniform on $(0, 1)$. The packet crosses Cartesian cells until its accumulated optical depth reaches $\tau_p$, is absorbed, and is re-emitted in situ from the local emissivity distribution. This is conceptually general and treats diffuse radiation naturally.

For the absorption-only ionizing band, MOCHII instead integrates the attenuated packet weight analytically through every crossed leaf:

$$\Delta J_E(\text{leaf}) = w L_{\text{pkt}} \frac{e^{-\tau_{\text{in}}} - e^{-\tau_{\text{out}}}}{\kappa(E)}, \tag{3}$$

with $\kappa(E)$ the leaf opacity at the packet's own photon energy $E$; the photoionization and photoheating integrals of every species are scored from this path at that energy (`src/raytrace_amr.f90`, `src/ion_score_mod.f90`). The packet is followed to the domain edge and contributes the expectation value of the absorbed path in each leaf. There is no sampled absorption location; statistical noise comes only from direction and photon-energy sampling. The analytic attenuation gate measured a mean photo-rate bias of +0.004%. The Owen-scrambled Sobol launch (`launch_sequence='sobol'`, the default since 2026-09-25) stratifies exactly those remaining launch variables: on the same gate the cell-wise photo-rate error then scales as $\sim 1/N$ rather than $1/\sqrt{N}$, with measured effective photon gains of 29× at $2^{17}$ packets and 222× at $2^{20}$, and the launch set becomes independent of the MPI task count. When dust scattering is enabled, MOCHII changes to a forced-first-interaction estimator, applies the dust albedo as a packet weight, samples an HG scattering direction, and uses Russian roulette at low weight.

Equation (3) is a substantial advantage for the present MOCHII target regime: ionizing radiation in H II regions is normally absorption-dominated. M$^3$ pays the variance of discrete absorption events even in that limit. Conversely, the M$^3$ packet cycle is more general for a diffuse spectrum containing many continuum and line channels, whereas the current MOCHII diffuse emitter carries named channels only: the H/He ground continua and excited cascades, the metal ground free–bound continua and the free–free, but no metal recombination lines and no metal captures into excited levels bound by more than 13.6 eV.

MAPPINGS V 5.2.1 has neither Monte Carlo estimator. Its zone integrator is deterministic and therefore essentially noise-free, but achieves that efficiency by imposing spherical or plane-parallel symmetry and an outward-only closure for the diffuse field.

## 6.2 Diffuse radiation and energy packets

In M$^3$, gas absorption converts the stellar packet into a diffuse packet immediately. Its new frequency is drawn from a local emissivity PDF containing recombination continua and lines, and its new direction is isotropic. This implements a full three-dimensional alternative to outward-only diffuse-field approximations [1].

MOCHII offers three controlled treatments of the recombination continuum (`par%recombination_continuum`). `'on_the_spot'` (case B) absorbs the ground-capture photons where they are made, and gives hydrogen its share of the He I and He II recombination photons that can ionize it (Osterbrock & Ferland 2006, eqs. 2.24–2.25 and sec. 2.5). `'transported'` (the default since 2026-09-26) uses case A and emits diffuse packets from an emission table rebuilt each iteration from the current state: the H II, He II and He III ground continua, the He I excited cascade (19.82 eV, 584 Å, two-photon) and the He III case-B cascade (free–bound into $n \geq 2$, 40.8-eV Ly$\alpha$, two-photon), the last two on by default in this mode, and since 2026-10-11 two further channels: the ground free–bound continuum of every

metal transition (channel 6, the Milne spectrum on the outer-shell part of the transport cross section, with ground-term weights from `data/atomic/ground_term_weights.txt`) and the free–free of $H^+$, $He^+$, $He^{2+}$ and the metal ions inside the band (channel 7). Measured on the HII40 geometry (2 iterations, not converged), channel 6 is $7.6 \times 10^{-4}$ of the diffuse luminosity (4.8% of channel 2) and channel 7 $4.5 \times 10^{-10}$; on a $10^5$ K blackbody $1.7 \times 10^{-3}$ and $2.1 \times 10^{-8}$. With `'on_the_spot'` channels 6 and 7 are not transported and their photons leave. `'escaping'` uses case A and lets the ground-capture photons leave. The packets use the same transport estimator as stellar packets. This split makes case-A/case-B bracketing and energy accounting transparent, but it is not yet the full local emissivity inventory sampled by M$^3$: metal recombination lines and metal captures into excited levels bound by more than 13.6 eV are not emitted. Free–free absorption is not in the opacity; its largest share of the H/He bound–free opacity over the band is $5.9 \times 10^{-17}$ at HII40.

## 6.3 Spectral discretization

The public M$^3$ compile-time ceiling is 10,240 frequency entries. The Monte Carlo loop launches $N$ packets for every frequency interval; packet energy is proportional to $L_\nu \Delta\nu / N$. This stratified sampling guarantees representation of weak spectral intervals and is attractive for detailed continua. Its cost scales approximately with $N_\nu N$ rather than with one total packet budget, and its large $J_\nu(x, y, z)$ and opacity arrays are major memory consumers.

MOCHII samples each packet's photon energy continuously from the source spectrum (`src/energy_sampler_mod.f90`: a piecewise-linear density with an exactly inverted CDF, adaptively refined for a Planck function), and the diffuse packets from their emission spectra, so the cross sections are evaluated at the packet's own energy and no band average enters the rates. Every packet of the ionizing band carries the same luminosity, and with `par%fuv_sampling = 'separate'` (the default) the FUV extension below 13.6 eV has its own packet budget on its own Sobol stream. The ionizing bins, still aligned on the H, He and active metal thresholds, now only resolve the diagnostic $J_\nu$ tally. Until the move to continuous energies the rates were taken from band-averaged bins, and the band discretization was a real lever on the helium and metal ionization. A line-rich diffuse spectrum is therefore no longer limited by a bin grid but by the channels the diffuse emitter carries.

## 6.4 Velocity fields and line transport

M$^3$ reads a three-dimensional velocity field and applies a projected Doppler shift to the packet frequency bin during continuum transport. MOCHII currently has no velocity-dependent band transport. This is a real M$^3$ advantage for bulk-flow effects on continuum opacity.

None of the three codes should yet be treated as a general resonance-line transfer solver. M$^3$ includes resonance emissivities but, in the published implementation, assigns resonance lines the same cross section as the continuum. This is not a frequency-redistributing resonant-scattering calculation. MOCHII explicitly treats its diagnostic lines as optically thin emissivities and leaves resonance transfer to a dedicated code such as LaRT. MAPPINGS V 5.2.1 applies geometry-dependent escape and attenuation approximations rather than multidimensional redistribution. For forbidden and recombination lines this is normally adequate; for Ly$\alpha$, C IV, Mg II, or line profiles in moving media, neither current path is sufficient.

# 7. Grid algorithms, resolution, and parallelism

## 7.1 Octree AMR versus a uniform Cartesian mesh

The primary spatial difference is adaptive resolution. M$^3$ accepts an arbitrary density distribution on a Cartesian mesh, but every cell at a given global resolution exists. The published HII20, HII40, and PN150 tests used $33^3$ and $55^3$ grids. Most integrated lines agreed within 10%, but I-front and boundary tracers such as [O I], [O II], and [S II] could differ by more than 10% [1]. This is the expected weakness of a uniform grid: the thin zone that most needs resolution occupies a small part of the volume.

MOCHII can ingest a pre-existing octree or rebuild the octree after an initial solution. A leaf is refined when it lies in the I-front or sees a large face-neighbor ionization jump. In the recorded gate, refinement from a level-5 start to level 7 produced 467,272 leaves instead of the 2,097,152 cells of a complete level-7 grid; the effective ionized radius agreed with the native level-7 result by 0.012% (2026-09-27, both runs stopped on the drift test). The refinement follows clump skins and distributed fronts in turbulent media, not merely the source position.

For problems that truly require a uniform mesh, MOCHII also has a raster-indexed Cartesian backend. It allocates no octree topology, neighbor table, center array, or leaf maps. The dedicated incremental Amanatides–Woo DDA precomputes $t_{\max}$ and $t_{\Delta}$ per ray and advances by comparisons and additions. In the recorded Strömgren test it was twice as fast as the octree traversal, while matching the solution to rounding. At $512^3$, omitting the tree and geometry arrays saves approximately 10 GB before physical fields are counted.

## 7.2 MPI decomposition and scaling trade-offs

M$^3$ spatially decomposes the Cartesian domain into equal blocks. Each rank stores the local thermal and ionization state, and packets crossing a block boundary are transferred through collective MPI operations. Spatial decomposition is attractive when no node can hold the full grid. It also introduces frequent synchronization: the current transport implementation collectively reduces large packet-state arrays containing flags, positions, directions, frequencies, and absorption counts. At large $N_\nu$, packet count, and MPI task count, communication can dominate.

MOCHII decomposes packets rather than space. The octree and leaf fields reside in MPI-3 shared memory, so ranks on one node share one copy; each node nevertheless holds a complete grid. Ranks transport independent packets and reduce the accumulated radiation tally at the end of a pass. This minimizes packet communication and is well matched to Monte Carlo transport, but the requirement of a full grid on each node ultimately limits the largest possible AMR problem. Thus the scaling comparison is not one-sided: MOCHII should normally scale more cleanly in packet number, whereas M$^3$ can distribute spatial memory more aggressively. MAPPINGS V 5.2.1 avoids the 3D memory problem entirely by advancing one 1D structure, which is a decisive throughput advantage only when its symmetry assumptions are physically defensible.

# 8. Ionization, thermal balance, and atomic data

## 8.1 Element and ion inventories

The local M$^3$ default atom file activates 16 elements: H, He, C, N, O, Ne, Na, Mg, Al, Si, S, Cl, Ar, Ca, Fe, and Ni. Static array dimensions allow up to 30 elements and 31 ion stages. The published description characterizes the parent MAPPINGS V system as containing 30 elements and roughly 80,000 cooling and recombination lines [1]. The distinction between the parent capacity and the default public data set should be retained when quoting these numbers. The new MAPPINGS V 5.2.1 audit resolves the parent-code side directly: `mapStd.prefs` is the 16-element setup and `mapFull.prefs` activates 30 elements from H through Zn. This full preference set, rather than the M$^3$ default, is the relevant breadth reference.

MOCHII currently treats H and He plus an eleven-element metal registry: C, N, O, Ne, S, Ar, Mg, Fe, Si, Cl, and Ca, each at stages I–VII (77 stages, 66 transitions, since 2026-10-11; 43 stages before), the stage scope of MOCASSIN v2. New ions are added through provenance-bearing element-rate files and the atoms of the selected excitation model. This is much easier to audit and extend than the global MAPPINGS tables, but the present inventory is less suitable for X-ray or coronal-ion problems, and a band above the terminal ionization potential of a present element (lowest Cl VII, 114.2 eV) is refused.

## 8.2 Photoionization, recombination, and charge exchange

MOCHII uses exact hydrogenic ground-state cross sections for H I and He II, the Verner et al. [9] fit for He I, and for the metals Verner's `phfit2` prescription in full: the outer-shell fit below the first inner-shell edge and the sum of the Verner & Yakovlev (1995) subshell fits above it, for the opacity, the rates and the heating (restored 2026-10-11; there is no option to keep the outer shell alone, and `par%metal_inner_shell_absorption` is refused). This brings the Mg $2p$, Ca $3p$ and Fe $3p$ edges into the 13.6–100 eV band. The product of every absorption is the next stage: the Mg I $2p$, Ca I $3p$ and Fe I–III $3p$ holes autoionize, so those absorptions should raise the stage by two and heat with the Auger electron, which the adjacent-stage solve cannot carry (a known defect; the stage matrix is planned as `par%auger_ionization`). The Ne III ground term takes the Nahar (2019) $R$-matrix cross section, resonances included, below 178.42 eV and `phfit2` above. Hydrogen and helium recombination (`par%recomb_model = 'detailed_balance'`, the default) takes the ground-state coefficient $\alpha_1$ from the Milne relation on the transport cross sections for all three ions; the case-B coefficient of $H^+$ and $He^{2+}$ comes from a zero-density $nl$-resolved cascade on the Storey & Hummer (1991) hydrogenic routines, with $\alpha_A = \alpha_B + \alpha_1$ (since 2026-09-26), and that of $He^+$ is the Badnell (2023) total minus $\alpha_1$; the Hui–Gnedin fits are retained as an option. Metal RR and DR are derived primarily from Badnell/CHIANTI v11.0.2 data, with $Fe^{3+}$ from Zhang et al. (2026, RR and DR). Charge exchange with H carries 45 reactions over all eleven elements (7 ionization, 38 recombination), every one also entering hydrogen's balance: the modern first-stage rates of Huang et al. (with $N^0/N^+$ from Kingdon & Ferland 1996 and $S^+$ resolved by product term from Zhao et al. 2005), the Barragan et al. (2006) tables for $N^{2+}$ and $O^{2+}$, the updated Kingdon–Ferland database distributed with CLOUDY (Ferland et al. 2013) for $N^{4+}$ and $Si^{2+}$, and the Kingdon–Ferland fits, tabulated over their published intervals, for the other stages up to $q = 4$, $Cl^{3+}$, $Cl^{4+}$, $Ca^{3+}$ and $Ca^{4+}$ included. For $q = 5, 6$ the Langevin upper limit (`par%langevin_charge_exchange`, 22 reactions)

is an option, off by default. Charge-exchange heating is on by default. He–H charge exchange enters the H/He balance (`par%helium_charge_exchange`, default on since 2026-09-24), with $He^+ + H$ taken since 2026-09-27 from Loreau et al. (2014) by reciprocity (non-radiative) and Stancil, Lepp & Dalgarno (1998) (radiative). Each generated file records its source, fit range, formula, and maximum fit error.

M$^3$ uses the established MAPPINGS compilation. The published code description identifies photoionization data from Seaton, Raymond, Osterbrock, and Gould, including Auger corrections, and separates hydrogenic from non-hydrogenic recombination. Much of this compilation is older than the MOCHII sources, but it spans more ions and has been exercised in many MAPPINGS applications. "Newer" is not automatically equivalent to "more accurate": line predictions can differ by several to tens of percent among reputable collision-strength and $A$-value compilations, especially for iron. The decisive test is therefore ion-by-ion comparison with direct database evaluation and observation, not vintage alone.

## 8.3 How the ionization system is actually solved: no code couples the elements

Comparisons of this family normally stop at the atomic data and the thermal solver, and leave the structure of the ionization solve as an assumption: that the established codes solve the ionization of every element as one coupled nonlinear system, and that MOCHII's element-by-element cascade is a reduction of that. Six codes were read in their own sources to settle it — MOCASSIN, CMACIONIZE, TORUS, IONIZE, COLT, and CLOUDY c25.00 — and *none of them does*. Every one solves one element at a time inside nested outer iterations, and five of the six use the same nearest-neighbor ratio chain MOCHII uses. The electron temperature is an outer root find in all five; it is never an unknown of the ionization system. The ratio chain is therefore the standard method of the field, not a MOCHII-specific approximation. CLOUDY departs from it on exactly two axes, both identified below, and neither of them is element coupling. COLT was added to the audit after its source became available and did not change the finding; its entry is Section 5.5 and its row is in Table 4. MAPPINGS V 5.2.1 and M$^3$ were not part of this audit, and no claim is made here about their solve structure.

In MOCASSIN, `ionBalance` (`update_mod.f90`:1321) forms the adjacent-stage ratio `ionRatio(elem,ion)` $= X_{i+1}/X_i$ (declared `:1356`, formed `:1495--1499`), accumulates it into cumulative products (`:1506--1517`), and closes each element with a single normalization scalar, `denominator(elem)`, dimensioned `(nElements)` (`:1348--1349`, formed `:1520--1527`, fractions `:1537--1544`). There is no ionization matrix in the tree at all: the only LU factorization, `luSlv/lured` (`emission_mod.f90`:2126–2164), is called from `:2091` inside `equilibrium` for the bound-level populations of the forbidden lines. The element solves are tied together only by an outer fixed-point recursion, `nIterateX` with `maxIterateX = 25` (`update_mod.f90`:84–85, recursive self-call `:1608`). CMACIONIZE writes the same chain out by hand, one `#ifdef` block for each element: `compute_ionization_states_metals` (`IonizationStateCalculator.cpp`:323–501, the scheme documented at `:280--310`), with carbon at `:392--403` as `C21 = jCp1/(ne*alphaC[0])`, `C31 = C32*C21`, `sumC_inv = 1./(1. + C21 + C31)`, and independent blocks for C (`:389--404`), N (`:406--434`), O (`:436--455`), Ne (`:457--471`) and S (`:473--500`); the metals are solved only after H/He has been frozen (`:116--185`). TORUS blocks its flat ion list by nuclear charge (`photoionAMR_mod.F90`:6859–6867), forms the same ratio (`:6880--6881`), chains it (`:6891`), renormalizes each element (`:6902--6904`), and under-relaxes the update by 0.6 (`:6853`, `:6914`). IONIZE is the extreme case: the chain is written out for carbon at `ionize2.f`:68–77 and as three links for nitrogen and sulfur at `:96--98`, each metal block reads only `ne`, `h0` and `he0`

from the already-converged H/He solution and never another metal's state, and there is no iteration on the metal fractions at all — one pass for each cell in each Monte Carlo iteration. Its only matrix, `SIMQ` in `SIMQ2.f`, again belongs to the five-level bound populations of the cooling function (`linecool1.f`:110). MOCHII does exactly what those four do: `species_fractions` (`src/species_mod.f90`:1363–1393) takes $T_e$, $n_e$, $n_{HI}$ and $n_{HII}$ as inputs and returns the chain product for one element.

Table 4: How the ionization system is solved in the six codes read in source for this subsection, with MOCHII as the last row. Entries describe the connected photoionization path, not what exists in the tree. "not audited" means the axis was not read for that code and no claim is made about it.

| Code | Ionization solve | Elements | Two-way charge exchange | Metals in $n_e$ | Metal opacity | Metal diffuse | $T_e$ solve |
|---|---|---|---|---|---|---|---|
| MOCASSIN | nearest-neighbor ratio chain | separate | no | yes, charge-weighted | yes, all shells | continua transported, lines discarded | outer secant, 25 passes |
| CMACIONIZE | chain, hand-written for each element | separate, metals after H/He | no | no | no | no metal channel | outer three-point power law, off by default |
| TORUS | chain, under-relaxed 0.6 | separate | no | yes, weight $Z-N$ | no; H I and He I only | not audited | outer Brent, under-relaxed |
| IONIZE | chain, one pass, no metal iteration | separate, metals after H/He | no | no | no | not audited | outer secant in log, from iteration 5 |
| CLOUDY c25.00 | $N \times N$ LU with iterative refinement | separate | yes, into hydrogen's own rows | yes, summed then Brent root find | yes, all shells | continua and lines, both re-ionize | outer Brent inside the pressure loop |
| COLT | chain as a recurrence, fixed point on $n_e$ damped 0.5 | separate | no; four partners, all one-way | yes, summed into $x_e$ | yes, every active ion | no metal channel; H/He free–free, free–bound and two-photon, opt-in | none in this tree; $T_e$ from the internal energy |
| MOCHII | chain for the metals; hydrogen's balance solved for its root in $u = \ln(n_{HII}/n_{HI})$ at each $n_e$ rather than substituted into | separate | **yes**, metal–H and He–H | yes, default on, damped fixed point | yes, all shells, default on | ground continua and free–free transported, no lines | outer bisection in $\log T_e$ |

CLOUDY's first difference is the matrix, and its reason is physical rather than numerical. It allocates a square matrix `xmat` of order `ion_range = IonHigh-IonLow+1` (`ion_solver.cpp`:70–72), assembles it in `fill_array()` (`:538--748`), and factorizes it at `:233` through `newton_step.cpp`:346–400 (`getrf_wrapper`/`getrs_wrapper`, with `nrefine = 3` rounds

of iterative refinement), one row being replaced by abundance conservation (`:883--889`). The matrix is deliberately not tridiagonal: the ionization block at `:727--736` runs the destination stage from `ion_from+1` up to `IonHigh` and fills arbitrary super-diagonal entries, because Auger multi-electron ejection removes several electrons in one event and therefore couples stage $i$ to $i+2, i+3, \ldots$ (`:661--668`). A nearest-neighbor chain cannot represent that, which is exactly why the MOCHII hard-spectrum plan of Section 16. replaces the chain with a small $n_{\rm stage} \times n_{\rm stage}$ matrix in its HS6 increment, at the point where an inner-shell channel opens. Two consequences are worth recording. The legacy chain solver is still in the file — `solveions()` at `:1317--1364`, with the tridiagonal matrix drawn in the comment at `:1290--1315` — and it is dead code, never called in c25. And the matrix buys CLOUDY nothing in element coupling: `ConvBase` loops `nelem` from `ipHYDROGEN` to `LIMELM` calling `ion_wrapper(nelem)` (`conv_base.cpp`:557–566), one solve for each element, inside an inner repeat loop of `nconv` = 3 passes (`:475`, `:539`).

The second difference is the one that matters physically for the codes compared here: CLOUDY's charge transfer is two-way in the bookkeeping. A reaction $X^{i+1}+\mathrm{H}^0 \rightarrow X^{i}+\mathrm{H}^+$ recombines the metal and ionizes the hydrogen in the same event, so it is a source and a sink on both sides. `iso_charge_transfer_update()` (`iso_ionize_recombine.cpp`:18–50), called at the top of every ion solve (`ion_solver.cpp`:68), accumulates hydrogen's charge transfer ionization and recombination totals over all heavier elements and all their stages, each weighted by the partner ion density (`:38--45`), with the source comments naming the symmetry outright ("charge transfer ionization of nelem1, recombination for other species"). Those totals then enter hydrogen's own matrix rows: `ction` at `ion_solver.cpp`:687 with its off-diagonal pair at `:696--697`, and `ctrec` at `:709` with `:723--724`; the H-like level matrix carries the same terms under `if(nelem==ipHYDROGEN)` (`iso_level.cpp`:121–124). Both sides read the *same* `CharExcIonOf`/`CharExcRecTo` arrays, and `ChargTranEval()` is called once in each outer loop precisely so that they do, as its own comment states: "charge transfer evaluation needs to be here so that same rate coefficient used for H ion and other recombination" (`conv_base.cpp`:498–502). CLOUDY then *checks* the balance instead of assuming it: `:578--583` forms the net rate `netion[nlo][nhi]` as the difference of the two directions, and a failure declares the zone unconverged with the reason "`CX inconsistency`" (`:611--620`). The honest caveat is that within a single LU the partner's density is a frozen coefficient, so exact charge conservation is reached by the outer iteration together with that check, not inside one matrix solve.

The other four codes are all one-way in that bookkeeping, and so is MOCHII. The evidence is specific in each case. In MOCASSIN the four-parameter table `chex(nElements,nstages,4)` (`update_mod.f90`:1358–1359) is populated for He through Fe at `:1392--1451`, fitted at `:1462--1463`, hard-zeroed outside 6000–50 000 K at `:1466`, and enters the *metal* denominator as `chex(elem,ion,1)*Hden*ionDen(H0)` (`:1498--1499`) — but `chex(1,...)` is never assigned anywhere in the tree, so it keeps the zero it was initialized with at `:1392`. Hydrogen's equation is the same generic line `:1495--1499` evaluated at `elem=1, ion=1`, which reduces to photoionization plus collisional ionization over $n_{\rm e}\alpha_{\rm tot}$ and contains no charge transfer term. Helium does carry one against $\mathrm{H}^0$ (`chex(2,1,:)`, `chex(2,2,:)`, set `:1395--1396`), but one-way, and with `deltaE_k(2,*) = 0` it has no reverse rate; the reverse rate is nonzero only for N I and O I, the only two entries ever set (`deltaE_k(7,1) = 10863.`, `deltaE_k(8,1) = 2205.` at `:1448--1449`, gated `:1475`). In CMACIONIZE, a search for `charge_transfer` returns no hit above line 501 of `IonizationStateCalculator.cpp`, that is, none inside the H/He solve (`:649--753`), whose only term beyond photoionization and radiative recombination is the helium Lyman-$\alpha$ on-the-spot term `ch2` (`:672--674`, used `:712`); `ChargeTransferRates.cpp`:49–52 and `:196--199`

even `cmac_error()` on a request for hydrogen with itself. In TORUS, `getChargeExchangeRecomb` holds data for N IV, O II and O III only (`photoion_utils_mod.F90`:840–868) and `getChargeExchangeIon` for N I and O I only (`:885--906`); both initialize to zero and fall through for every other nuclear charge (`:838`, `:883`), and the hydrogen balance calls those same routines with `parentIon = grid%ion(2)` and `grid%ion(1)` (`photoionAMR_mod.F90`:6872–6878), which return zero, so hydrogen reduces to photoionization over recombination (`:6880--6881`). In IONIZE, `ct1.f` is the Kingdon & Ferland package itself (`HCTRecom` at `:3`, `HCTIon` at `:37`, `block data ctdata` at `:72`, covering $Z = 2$–30) and it is called in the metal denominators (O `ionize2.f`:136–138, N `:83--85`), but `h0find.f` has no charge transfer argument in its interface (`:1`), its stated equation set (`:6--13`) contains only hydrogen photoionization, recombination and the helium 2s–2p resonance term, and neither `HCTIon` nor `HCTRecom` appears in the file. MOCHII shared this until 2026-07-31: `species_fractions` added the two terms to the metal rates while `solve_ion_cell` carried none.

Physically the omission means the same thing in all four remaining codes: charge is not conserved in the hydrogen-plus-metal pair. Every charge transfer recombination of a metal ion that these codes count also ionizes a hydrogen atom, and that hydrogen ionization is never entered.

MOCHII now enters it (`par%charge_exchange = 'two_way'`, the default), so on this axis it sits with CLOUDY rather than with the Monte Carlo codes; `'metal_only'` reproduces the earlier bookkeeping bit for bit. The correction turns out to be small on the Lexington benchmarks, and the reason is structural rather than one of abundance: where the near-resonant $O^0 + H^+$ pair locks oxygen to hydrogen the two terms cancel identically in hydrogen's balance, so only the departure from that lock survives. Measured with a single switch changed, $\langle T[N_p N_e] \rangle$ moves −1.41 K at HII40 and −0.13 K at HII20 and every front by under 0.001 pc — inside the tolerance those runs were converged to by factors of 58 and 533. The same near-cancellation makes the substitution map marginally stable in partially neutral gas, which is why MOCHII solves hydrogen's balance for its root there rather than substituting into it.

On two axes MOCHII is ahead of three of the four chain codes. The first is opacity. MOCHII attenuates the ionizing field with the metal cross sections by default (`par%ion_metal_abs = .true.`); CMACIONIZE, TORUS and IONIZE do not. CMACIONIZE's `get_optical_depth` returns $ds\,n\,[\sigma(H^0)x(H^0) + \sigma_{\mathrm{He,corr}}x(He^0)]$ and nothing else (`DensityGrid.hpp`:117–140), with the task-based path identical (`DensitySubGrid.hpp`:557–583), while mean-intensity integrals are nevertheless accumulated for every ion including the metals (`DensityGrid.hpp`:162–181). TORUS calls `phfit2` exactly twice, for H I and He I — not even He II (`amr_mod.F90`:16301–16309) — and forms `kappaAbsGas = (kappaH + kappaHe)` (`:16311`), which is what sets the packet's optical depth in the cell (`photoionAMR_mod.F90`:4762, `:4741--4743`); its metal cross sections (`ion_mod.F90`:172) are consumed only to build `photoIonCoeff` (`photoionAMR_mod.F90`:6688–6692). IONIZE carries only $n_{\mathrm{HI}}\sigma_{\mathrm{H}} + A_{\mathrm{He}}n_{\mathrm{HeI}}\sigma_{\mathrm{He}}$ in `tauint.f`:159–161 and in the sub-cell step-back at `:169--171`, while computing the Verner metal cross sections at every frequency (`opacset.f`:20–55) and accumulating them into the metal photoionization rates in the same loop (`tauint.f`:194–216). In all three the metal photoionizations are counted as rates while the photons are never removed from the beam: they are sinks that consume no ionizing budget. MOCASSIN and CLOUDY do remove them. `addOpacity` (`ionization_mod.f90`:349, called `:127` from `ionizationDriver` and driven cell by cell from `iteration_mod.f90`:142) loops all metals and all shells (`:409--411`) into `putOpacity` (`:430--439`, `:476`), and that `grid%opacity` is what transport reads (`photon_mod.f90`:1390, `:1442`); CLOUDY does the same through `opacity_addtotal.cpp`:284–291 and `opacity_add1element.cpp`:41–62 into `opac.opacity_abs[]`, which attenuates the beam in `rt_continuum.cpp`:182–197.

The size of that omission was measured for this memo and it is not small. Switching `par%ion_metal_abs` off at HII40 moves $r(\mathrm{He}^0 = 0.5)$ inward by 0.615 pc: the stages with thresholds between 21 and 40 eV (Ne I 21.56, S II 23.33, C II 24.38, N II 29.60, S III 34.83, O II 35.12 eV) preferentially remove the softest helium-ionizing photons, while $\sigma(He\,\mathrm{I})/\sigma(H\,\mathrm{I})$ rises from 6.1 at 25 eV to 19.9 at 95 eV. The response is asymmetric and saturates upward: 0.615 pc for a unit of metal opacity removed against 0.103 pc for a unit added.

The second axis is $n_\mathrm{e}$. CMACIONIZE and IONIZE do not count metal electrons at all: `ne = ntot*(1.-h0 +AHe*(1.-he0))` (`IonizationStateCalculator.cpp`:134, identically `TemperatureCalculator.cpp`:275, whose comment at `:273--274` says "we neglect free electrons coming from ionization of coolants", the same assumption being documented at `IonizationStateCalculator.cpp`:576–586), and `ne=n*(1.-h0+AHe*(1.-he0))` (`ioneng.f`:45, also `ionize2.f`:54), which IONIZE states as a deliberate approximation at `tbal1.f`:39–41. MOCASSIN sums the charge-weighted contribution of every switched-on element (`update_mod.f90`:1564–1577, written back `:1592--1594`), with the caveats that ions below a fraction of $10^{-10}$ are silently dropped and stages above `nstages` cannot contribute at all; TORUS does the same with weight $Z-N$ (`ion_mod.F90`:1418–1421) and recomputes it after every ionization update (`photoionAMR_mod.F90`:6916, `:5941`). In CLOUDY, $n_\mathrm{e}$ is not an unknown of the matrix either: `eden_sum.cpp`:51–62 builds `dense.EdenTrue` as a diagnostic sum over all 30 elements plus deuterium, molecular and grain electrons (`:74`, `:78`), and the electron density actually used changes in exactly one place, `EdenError()` (`conv_eden_ioniz.cpp`:271–276), as the residual of a van Wijngaarden–Dekker–Brent root find (`:44--150`) each of whose residuals re-solves the ionization of every element through `ConvIoniz()` (`:283--288`).

MOCHII's metal contribution to $n_\mathrm{e}$ is optional (`par%metal_ne`, on by default since 2026-07-31, and since 2026-09-25 on whenever the input sets a metal abundance and leaves it out), and it is solved as a scalar fixed point rather than evaluated once: `charge_neutrality_ne` (`src/ion_balance_mod.f90`:968–986) solves

$$n_\mathrm{e} = n_\mathrm{e}(\mathrm{H},\mathrm{He}) + n_\mathrm{H} \sum_\mathrm{el} A_\mathrm{el} \sum_i (i-1)\, f_i(n_\mathrm{e}) \tag{4}$$

with the same damped iteration and tolerance as the H/He closure. The reason is recorded at `:950--953` and is worth restating, because the cheaper choice is wrong for a specific physical reason rather than merely inaccurate: evaluating the metal sum once at the H/He-only electron density is the $n_\mathrm{e} \to 0$ limit beyond the front, where the H/He term vanishes, nothing recombines, and every low-ionization-potential metal comes out fully ionized.

On this axis MOCHII was behind until 2026-10-11: its metal recombination continua were not transported. They now are, as diffuse channel 6 of `src/diffuse_mod.f90` (ground captures of every metal transition, the Milne relation on the outer-shell part of the transport cross section), and the free–free of every ion inside the band as channel 7. On the HII40 geometry (2 iterations, not converged) channel 6 is 4.8% of diffuse channel 2; an earlier estimate in this memo, 0.29%, is superseded by that transported measurement. Metal recombination lines remain untransported, where CLOUDY feeds them back. MOCASSIN builds `gammaHeavies` in `fb_ff` over `do elem = 3, 19` (`emission_mod.f90`:474, its own comment at `:472--473` noting that $Z > 19$ is neglected) from the Milne relation with the true subshell cross section (`:519--521`), folds it into the continuum (`:126--128`) and into `sumDiffuseHI` (`:983`) and the cumulative diffuse PDF `recPDF` (`:1175--1196`), which `newPhotonPacket` samples (`photon_mod.f90`:958) and propagates (`:461--466`). CLOUDY goes further and feeds both continua and lines back into the field, where they genuinely re-ionize: metal recombination continua are emitted as diffuse radiation from the stored Milne-inverted metal

cross sections (`rt_diffuse.cpp`:196–270), metal recombination Lyman-$\alpha$ and Balmer photons enter the on-the-spot field (`rt_ots.cpp`:362–406), both are added into `rfield.SummedCon` (`:588--592`), and `SummedCon` is exactly what `GammaK/GammaBn` integrate against the cross sections (`cont_gammas.cpp`:125, `:143`, `:409`) to produce `ionbal.Photo Rate_Shell` (`ion_photo.cpp`:100–110) — feedback onto hydrogen included.

Finally, thirteen properties of the reference codes surfaced in this audit and in the audit of 2026-10-11 (`md/ATOMIC_STAGE_EXPANSION_AUDIT_2026-10-11.md`, Section 5; the MOCASSIN line numbers below are those of the 2.02.73.2 tree, read) that change what a comparison against them means, and that a reader using them as validation references has to know about:

- MOCASSIN substitutes *oxygen's* dielectronic recombination rate wherever an element's own rate is missing: `if (diRec(elem,ion) == 0.)diRec(elem,ion) = diRec(8,ion)` (`update_mod.f90`:1648).
- MOCASSIN has no dielectronic recombination for helium: the loop that adds it starts at `elem = 3` (`update_mod.f90`:1644–1649).
- MOCASSIN's metal line cooling sums the forbidden-line array over levels 1–10 only (`do j = 1, 10`, `do k = 1, 10`, `update_mod.f90`:1103–1111), whatever the size of the atom.
- Fe II never enters MOCASSIN's cooling. Its populations are solved into the separate `forbiddenLinesLarge` (`update_mod.f90`:1277–1293), and the cooling sum reads `forbiddenLines` alone.
- MOCASSIN's $He^+$ recombination cooling uses the hydrogen coefficient: the $\beta_{rec}$ polynomial at `update_mod.f90`:1059–1064 is the $H^+$ fit of `:989--993`, evaluated at the unscaled temperature.
- MOCASSIN's free–free opacity is computed in the first frequency bin only. For every later bin `FFOpacity(i)` is set to zero when the previous bin is below $10^{-35}$ and is otherwise never assigned (`ionization_mod.f90`:368–390), so it keeps the zero of its initialization (`:111`).
- MOCASSIN's ionization convergence test examines $H^0$, $He^0$ and $He^+$ only (`:1598--1604`, `limit = 0.01` at `:1337`). No metal ion fraction is ever tested, so a converged MOCASSIN zone carries no statement at all about its metal ionization.
- MOCASSIN discards its metal line photons in production. The choice between a line and the continuum is made (`photon_mod.f90`:925), but the line identity is thrown away with `nuP = 0` outside `lgDebug` (`:926--941`) and the packet is dumped into `linePackets` and terminated without transport (`:468--475`). The line PDF is built only under `lgDebug` (`emission_mod.f90`:1209, metals `:1251--1289`), `grid%Jdif` is accumulated only under `lgDebug` (`photon_mod.f90`:1491–1493, `:1730--1732`) with everything otherwise landing in `Jste` (`:1488--1489`, `:1727--1728`), and `nPhotoDif` enters the ion balance only in the `lgDebug` branch (`update_mod.f90`:1489 versus `:1495`). `lgDebug` defaults to `.false.` (`set_input_mod.f90`:44).
- MOCASSIN truncates every element at `nstages = 7` (`set_input_mod.f90`:82) while carrying `nElements = 30` (`constants_mod.f90`:54), so iron stops at Fe VII.
- TORUS creates all four of its "Ar I–IV" ions with $n = 18$: all four are neutral argon (`ion_mod.F90`:331–340).

- CMACIONIZE's argon exists only as an enumeration. `ElementNames.hpp`:142–151 declares the ion names, no solver references them, and `HAS_ARGON` is defined only under `ADDITIONAL_COOLANTS` (`:46--49`).
- TORUS omits He II from the opacity, as above, so its helium-ionizing continuum is attenuated by neutral helium alone.
- IONIZE runs effectively isothermal for its first four iterations. The thermal solve is entered only `if(iter.ge.5)` (`itbal.f`:27–41), and before that `ionize2.f`:30 takes the temperature as given, which is the `Tconst = 8000.` seed of the shipped deck (`input.params`:6, passed at `mcionize.f`:115).

## 8.4 Electron distributions and secondary processes

Both M$^3$ and MAPPINGS V 5.2.1 support either a Maxwellian or a $\kappa$ electron energy distribution, with rate corrections supplied by KAPPA data. This is useful when exploring non-Maxwellian interpretations of nebular diagnostics. MOCHII assumes a Maxwellian distribution.

All three codes include secondary-ionization concepts. MOCHII optionally partitions the energy of H and He photoelectrons above 30 eV among heat and secondary H/He ionizations, with the Shull–van Steenberg (1985) ionization budget carried to lower energies by the Dalgarno, Yan & Liu (1999) mean energies per ion pair and divided among the neutral targets present. The MAPPINGS ion-balance routines call a secondary-ionization module and also include cosmic-ray ionization/heating. The MAPPINGS-derived codes therefore have the broader general-plasma framework. A Compton routine is present in both source trees, but no call from either examined main cooling path was found; it should not be counted as active physics without a dedicated test.

## 8.5 The critical difference: density-dependent metal cooling

This was the most important gas-physics advantage of both MAPPINGS-derived codes over MOCHII until 2026-07-26, and what remains of it is narrow (end of this subsection). Their thermal solvers call fine-structure, intercombination, multilevel, and iron line modules during every evaluation of the net cooling function. Level populations and collisional de-excitation therefore affect the equilibrium temperature when $n_e$ approaches or exceeds a transition's critical density. MOCHII deliberately separates two products:

- Tier-1 line cooling, evaluated cheaply inside the thermal bisection: $\Lambda(T, n_e)$ tables built from the $n$-level solve at the local $n_e$ (the default since 2026-07-26), or the analytic fits of the optically thin low-density limit as an option; and
- Tier-2 $n$-level statistical-equilibrium data, evaluated on the converged state to produce density-dependent diagnostic emissivities.

The output line ratios therefore respond to density, and since 2026-07-26 so does the thermal loop: `par%cooling_model = 'local_ne'` (the default) takes the line cooling from the $n$-level populations at the local $n_e$, tabulated as $\Lambda(T, n_e)$ for each ion, and the low-density fits (to the balance at $n_e = 10^{-6}$ cm$^{-3}$) remain as the `'low_density'` option. At the Lexington densities that is what brings the benchmark temperatures onto CLOUDY's (MOCHII physics document, the Lexington benchmark section). What M$^3$ still does that MOCHII does not is to solve the populations inside the thermal iteration for every ion rather than read a table computed from the same solver.

### 8.6 Thermal processes and equilibrium solution

All three codes solve ionization and temperature under the current radiation field, iterating the spatial solution where required. MOCHII brackets the root of heating minus cooling in $\log T_e$ and re-solves the H/He and metal ionization at every trial temperature. Its current budget includes photoionization heating, H/He recombination and collisional losses, free-free emission with a tabulated Gaunt integral (H/He and, by default, the trace metals with their net ion charge), metal-line cooling for every stage with an atom, metal photoheating with metal recombination and collisional-ionization losses (`par%metal_heat`, on when metals are enabled), and, on by default (restored 2026-10-11 after being off in the tree of October 9–11), metal collisional ionization (Voronov 1997), charge-exchange heating, atomic-hydrogen impact on the C I, C II and O I fine structure, and the He I $2^3$S population coupling. With dust it adds the Weingartner & Draine (2001) grain photoelectric heating and the cooling by charged-particle capture onto grains (`par%grain_pe`), and the Hollenbach & McKee (1979) exchange of neutral H and He with the grains (`par%gas_grain_energy_exchange`), both on with dust; secondary ionization remains an option (`src/thermal_mod.f90`, `src/cooling_mod.f90`).

The MAPPINGS thermal solver used by MAPPINGS V 5.2.1 and inherited by M$^3$ includes a broader traditional plasma budget: hydrogen and helium line cooling, fine/intercombination/resonance and multilevel metal cooling, free-free, collisional ionization, recombination, photoionization heating, charge-exchange energy exchange, cosmic rays, grain/PAH photoelectric terms, and grain collisional cooling. The microturbulent dissipation block in the examined 5.2.1 `cool.f` is commented out and is therefore not listed as active. This breadth and the direct multilevel coupling are major MAPPINGS strengths.

## 9. Dust, PAHs, and PDR physics

### 9.1 Where the MAPPINGS microphysics is stronger

The MAPPINGS grain microphysics is detailed. The local source tree carries graphite and silicate size distributions, depletion bookkeeping, grain charge balance, electron and ion sticking, grain photoelectric heating, collisional grain cooling, PAH charge states, and optional thermal or ionization-parameter-dependent grain destruction. MOCHII has, since 2026-10-11, replaced its Bakes–Tielens (1994) fit with the Weingartner & Draine (2001) calculation for the size bins of its own grain model: the detailed-balance charge distribution with photodetachment, the photoelectric heating in the band field the transport has in each leaf, and the cooling by captured electrons and ions (`src/grain_photoelectric_mod.f90`); and it adds the Hollenbach & McKee (1979) exchange of neutral H and He with the grains (`src/gas_grain_collision_mod.f90`). PAH grains count in the heating with the weight $x_{HI} + f_{ion,PAH}\, x_{HII}$, $f_{ion,PAH} = 0$ by default (destroyed inside the ionized gas). What MAPPINGS still has beyond that is depletion bookkeeping and grain destruction, which MOCHII does not carry. MOCHII also has no grain-assisted recombination in its ionization balance (whether MAPPINGS connects one was not checked).

In MAPPINGS V 5.2.1, this layer is connected to the P6/P7 zone integration and to `dusttemp`, which computes a Draine–Li stochastic temperature distribution and an IR photon field. It is therefore a complete 1D dust emission path, not merely a library capability. The distinction below is specific to the public M$^3$ three-dimensional branch.

## 9.2 Where the three-dimensional M$^3$ dust path is limited

The public `montph7_dust` branch includes dust in the extinction and terminates a packet when the selected interaction is grain absorption. The dust data contain absorption, scattering, and asymmetry information, but the examined 3D branch computes an absorption fraction and does not sample a new dust-scattering direction. The dedicated `dusttemp` routine is called from the one-dimensional `photo6/7` family, not from the public 3D driver. Consequently, the rich MAPPINGS grain microphysics should not be conflated with a complete 3D dust radiative transfer and emergent-IR calculation.
This does not imply that dust is absent from M$^3$: dust opacity, depletion, grain heating/cooling, and PAH processes affect the gas. It means that the public path does not presently expose a validated chain equivalent to MOCHII's dust scattering, absorbed-energy ledger, stochastic emission, and observer maps.

## 9.3 The MoCHII dust chain

MOCHII couples dust to the same FUV/EUV radiation field that sets the gas state:

1. Gas and dust compete for ionizing photons using absorption cross sections evaluated at each packet's photon energy. The dusty Strömgren gate reproduces the one-dimensional radius to +0.69% and the classic radius reduction $R/R_0 = 0.735$.

2. With scattering enabled, the extinction includes the dust scattering coefficient, a forced first event is sampled, and the packet follows an HG direction with an albedo weight. The dusty-sphere result lies between the pure-absorption and extinction-as-absorption bounds.

3. The live dust model updates the dust abundance from the computed neutral fraction. A separate PAH survival fraction permits destruction inside ionized gas and produces a PAH-bright I-front ring.

4. The absorbed FUV/EUV power gives a leaf equilibrium mixture temperature and an energy-conserving IR spectrum.

5. SEDust supplies astrodust, DL07, MRN, or Zubko stochastic emission (MRN has no PAH component). The final normalization enforces $\int L_\lambda \, d\lambda = \sum_{\text{leaf}} H_{\text{dust}} V$; the smoke test closes at unity and resolves the usual PAH bands.

6. Leaf SEDs at selected wavelengths generate spatially resolved 8, 24, 70, and 160-$\mu$m maps.

For dusty three-dimensional H II regions, this end-to-end chain is the clearest overall physical advantage of MOCHII.

## 9.4 PDR scope

MOCHII extends the band below 13.6 eV so that low-threshold metals and dust see the FUV field beyond the H I-front; left out of the input, the extension is on (from 4.78 eV) whenever He I $2^3$S, a metal with a neutral edge below 13.598 eV, or dust is present. Electrons from the metal cascade, metal photoheating, the Weingartner–Draine grain photoelectric heating with capture cooling (with dust and the FUV band), the Hollenbach–McKee gas–grain exchange of the neutral atoms (with dust), and the neutral-gas coolant, atomic-hydrogen impact on the ground-term fine structure of C I, C II and O I (all of [C I] 609/370 $\mu$m, [C II] 158 $\mu$m and [O I] 63/145 $\mu$m), are all on by default; the last is solved with

the electrons in one statistical equilibrium. This is an atomic PDR skin, not a complete PDR code. $H_2$/CO chemistry, molecular self-shielding, molecular cooling, line transfer, and grain-assisted recombination remain outside the model. The MAPPINGS trees likewise contain more grain and general plasma physics than a minimal H II solver, but the present comparison did not identify a full molecular $H_2$/CO network in either the 5.2.1 photoionization path or the public M$^3$ 3D driver. None of the three codes should be selected as a substitute for a dedicated molecular PDR code without additional chemistry.

# 10. Emission lines, continua, and synthetic observations

## 10.1 Line inventories and emissivity accuracy

The MAPPINGS lineage was designed to predict a very broad line spectrum. The M$^3$ cooling and output path includes resonance, intercombination, forbidden, fine-structure, hydrogenic, helium, heavy recombination, and extensive iron-line data. Its five-level and multilevel populations are already part of the thermal solution. This makes M$^3$ attractive for integrated diagnostic grids and hard nebulae for which a large fraction of the periodic table and many ion stages matter. MAPPINGS V 5.2.1 is the strongest version of that argument when 1D geometry is acceptable: the full preference set covers 30 elements, and the deterministic zone solver can afford a detailed spectrum without Monte Carlo sampling noise. M$^3$ trades some parent-code recency and 1D special modes for arbitrary Cartesian geometry.

MOCHII has a narrower but highly auditable diagnostic layer. Every stage with an atom in the selected excitation model is solved by the same statistical-equilibrium code (LU with partial pivoting and one step of iterative refinement) on the full CHIANTI 11.0.2 atoms by default (70 atoms, 6629 levels; `par%metal_excitation_model = 'chianti'`), with the effective collision strengths held at the end of their tabulated range beyond it; the line-cooling tables of the thermal loop are built with the same solver on the same atoms. H I and He II recombination lines use Storey–Hummer case-A/B tables. The nebular continuum includes free-bound plus free-free and two-photon components. On an identical state, the Fortran line luminosities match the independent Python solver to at least four significant figures. The limitation is coverage, not the internal emissivity algebra.

## 10.2 Hydrogen recombination lines

Hydrogen recombination lines deserve their own comparison, for two reasons. They are the normalization of every published line list, so an error in H$\beta$ propagates into every ratio quoted against it; and they are the one emission process every code here computes, which makes them the cleanest available cross-section through the design choices of the family. CLOUDY c25.00 is carried in this subsection as well, because it answers the question in a different category from the other six. The seven implementations fall into four method classes.

*Table interpolation in* $(T_e, n_e)$. MOCHII reads the Storey & Hummer (1995) case-A and case-B tables (11 temperatures $\times$ 13 densities for case B, 10 $\times$ 9 for case A) and interpolates bilinearly in $(\log T_e, \log n_e)$, clamped at the grid edges, for seven lines (H$\alpha$, H$\beta$, H$\gamma$, H$\delta$, Pa$\alpha$, Pa$\beta$, Br$\gamma$). MAPPINGS V 5.2.1 and M$^3$ read the same Storey–Hummer calculation in a richer layout: an absolute H$\beta$ emissivity on 12 temperatures $\times$ 9 densities for *each* case, spline-interpolated in $\log T_e$, times 90 line ratios (15 lines $\times$ 6 series) on their own $10 \times 9$ grid (`HRECDAT.txt`, `HRECRATA.txt`, `HRECRATB.txt`;

`hydro.f`:378–459). The density grid reaches $10^{10}\,\mathrm{cm}^{-3}$, against $10^{14}$ in MOCHII's case-B file and $10^{10}$ in its case-A file. MOCASSIN's *output* path is also a table interpolation, of `e1bx.d` — the same Storey–Hummer file MOCHII parses — by fourth-order Lagrange in $\log n_\mathrm{e}$ and $\sqrt{T_\mathrm{e}}$, up to $n_\mathrm{up} = 30$.

*Fixed ratios times a fitted H$\beta$.* MOCASSIN's *iteration* path reads `r1b0100.dat`, whose footer records what it is: `!Te=1.e4 Ne=1.e2 Case B`. The ratios carry no temperature or density axis, so H$\alpha$/H$\beta$ is fixed at the tabulated 2.86 whatever the local state, and the only temperature dependence of any H line is the H$\beta$ normalization. TORUS does the same thing with the same numbers: a ratio table stated in its own comment to be "for $T = 10^4$ K and $n = 10^2\,\mathrm{cm}^{-3}$" times an H$\beta$ power law (`photoionAMR_mod.F90`:8022–8043). The consequence is worth naming, because it is a physical one: in both codes *the Balmer decrement cannot respond to temperature at all*. Against Storey–Hummer at $n_\mathrm{e} = 100\,\mathrm{cm}^{-3}$ the frozen 2.86 is −5.9% at 5000 K, right at $10^4$ K, +4.1% at $2 \times 10^4$ K and +5.9% at $3 \times 10^4$ K.

*Analytic power-law fits.* IONIZE and CMACIONIZE carry *numerically identical* coefficients — CMACIONIZE is the C++ rewrite, and this is where that shows most plainly:

$$\frac{4\pi j(\mathrm{H}\beta)}{n_\mathrm{e} n_\mathrm{p}} = 1.24 \times 10^{-25}\, T_4^{-0.878}, \qquad \frac{4\pi j(\mathrm{H}\alpha)}{n_\mathrm{e} n_\mathrm{p}} = 2.87 \times 1.24 \times 10^{-25}\, T_4^{-0.938} \tag{5}$$

in $\mathrm{erg\,cm^3\,s^{-1}}$ (`lines.f`:219–221; `src/EmissivityCalculator.cpp`:224–229), the H$\beta$ coefficient attributed in both to Storey & Hummer (1995) and the 2.87 to Osterbrock's case-B table. Since the two exponents differ, the ratio drifts as $T_4^{-0.060}$ rather than staying at 2.87, which is what makes the fit work as well as it does.

*A solved level population.* CLOUDY has no emissivity table for hydrogen at all. Its hydrogen line data are reduced Einstein $A$ coefficients for $n' \to n$ transitions (`data/hydro_tpn_n100.dat`, to $n = 100$ by default, with an $n = 2000$ set shipped alongside), and the line emission is whatever the solved H-like iso-sequence level populations give. The default structure is ten resolved levels (`iso.cpp`:192), raised by the `database h-like resolved/collapsed levels` command. Nothing is interpolated, so the low-density limit is a result of the solve rather than the edge of a grid, and the same applies at the top: CLOUDY computes through $n_\mathrm{e} = 10^{13}\,\mathrm{cm}^{-3}$, where a table-reading code has left its data behind. The price is that the answer is only as good as the level structure, which its own test suite states explicitly (below).

The three tabulation-derived families can be compared against one common standard, because MOCHII's parse of `e1bx.d` is the same Storey–Hummer case-B table that four of the six table- or fit-based codes ultimately cite. Table 5 evaluates each H$\beta$ prescription against it.

Two readings follow. The fits are good: over the range an H II region actually occupies, both fit families reproduce the table they are fitted to within a few percent, so the analytic route is a representational choice rather than an accuracy defect. And MOCASSIN's reported absolute H$\beta$ is systematically 4.6–5.2% below the very table the same routine interpolates for its ratios, across the whole nebular range. That matters for this project specifically: MOCHII takes H$\beta$ from the table and MOCASSIN from a fit below it, so a few percentage points of any $L$(H$\beta$) comparison between the two codes is a coefficient artifact rather than physics. Line *ratios* are immune, because MOCASSIN divides by its own H$\beta$ before writing them.

**What the interpolation scheme costs.** Reading a table exactly at its nodes is not the same as agreeing between them, and the three codes that interpolate the Storey–Hummer grid do it three different ways. MOCHII is bilinear in $(\log T_\mathrm{e}, \log n_\mathrm{e})$ on $\log j$, i.e. a power law within each cell (`sh95_mod.f90`:161–174). MAPPINGS V 5.2.1 and M$^3$ take a

Table 5: Each code's H$\beta$ emissivity divided by the Storey & Hummer (1995) case-B table value at $n_e = 100\,\mathrm{cm}^{-3}$, computed from the coefficients in the sources. IONIZE and CMACIONIZE share Eq. (5); TORUS and MOCASSIN's iteration path share $10^{-0.870\log_{10}T_e+3.57}$; MOCASSIN's output path uses $2530\,T_e^{-0.833}$ while the accurate fit sits commented out one line above it (`output_mod.f90`:2060–2063). All six temperatures listed are nodes of the table, so the three codes that interpolate it return the table value there and the interpolation scheme does not enter; what it costs between nodes is measured in Table 6.

| $T_e$ / K | MOCHII, MAPPINGS V 5.2.1, M$^3$ | IONIZE, CMACIONIZE | TORUS, MOCASSIN (iteration) | MOCASSIN (output) |
|---|---|---|---|---|
| 5 000 | 1.000 | 1.036 | 1.022 | 0.954 |
| 7 500 | 1.000 | 1.011 | 1.001 | 0.948 |
| 10 000 | 1.000 | 1.004 | 0.996 | 0.954 |
| 15 000 | 1.000 | 1.010 | 1.005 | 0.977 |
| 20 000 | 1.000 | 1.026 | 1.023 | 1.005 |
| 30 000 | 1.000 | 1.065 | 1.065 | 1.062 |

natural cubic spline in $\log_{10}T_e$ of $\log_{10}j$ and interpolate linearly in $\log n_e$ (`hydro.f`:255–265, 435–459). MOCASSIN's output path uses a five-point Lagrange polynomial in $\sqrt{T_e}$ and $\log_{10}n_e$ applied to $\log(j\sqrt{T_e})$, dividing the smoothing factor out afterwards (`output_mod.f90`:2234–2366, `data maxv/4/ ni/2,3,4,5,6/`) — the scheme Storey & Hummer's own interpolation program uses.

Table 6 evaluates the three on the same case-B table. The spread is identically zero at the grid nodes and reaches a few tenths of a percent between them, largest where the grid is coarsest and at the high densities where the emissivity turns over. Two observations follow. The choice of scheme is not a material source of disagreement between these codes: at 0.1–0.3% it is an order of magnitude below the offset of MOCASSIN's fitted H$\beta$ (Table 5) and below the 0.4–3.6% of the analytic fits. But the error of the low-order scheme is *one-sided*, which is the property that matters: the chord lies below the curve, so MOCHII's bilinear form is low at every off-node point rather than scattering about zero, and integrating over a model cannot average it away.

Because MOCHII's $L$(H$\beta$) then sat below the published benchmark values, that bias was the natural suspect, and it was measured on the converged models of that time rather than estimated (the numbers of this paragraph and the next are from those runs, before 2026-09-25, and were not re-measured; on the converged run `m13_restoration` of 2026-10-11 $L$(H$\beta$) is $2.025\times10^{37}$ and $4.85\times10^{36}$ erg s$^{-1}$, inside the published 2.01–2.10 and 4.83–5.04, at their low ends; `m11` gave $2.02\times10^{37}$ and $4.85\times10^{36}$). Re-evaluating the same converged state — leaf by leaf, at each leaf's own $(T_e, n_e)$ — with the two higher-order schemes in place of the bilinear one changes the volume-integrated $L$(H$\beta$) by

$$-0.061\%\ \text{(HII40)}, \qquad -0.126\%\ \text{(HII20)}, \tag{6}$$

identically against the spline and against the Lagrange form. The bias is indeed systematic and not statistical: of the 136 168 emitting leaves at HII40 *every one* is biased in the same direction, with a median of –0.065% and a worst case of –0.217%; at HII20, 93.5% are, with a median of –0.130%. HII20 carries twice the bias of HII40 because its

temperature falls in the wider 5000–7500 K cell of the table where the chord error is largest. The Balmer decrement is less sensitive still, $\leq 0.05\%$, since the bias is common to the two lines and divides out of the ratio.

Set against the 3–5% deficit those runs had, the interpolation algorithm therefore accounted for one part in fifty at HII40 and one part in thirty at HII20. The remainder is the photon budget: the converged models return 0.950 and 0.970 of $Q_\mathrm{H}$ as hydrogen recombinations, with helium taking $0.112\,Q_\mathrm{H}$ at HII40, and $L(\mathrm{H}\beta)$ follows that budget rather than the emissivity table. Removing the one-sided bias — a spline or the Storey–Hummer $\sqrt{T_\mathrm{e}}$-smoothed form in place of the chord — is worth doing because a systematic error should not be left in a normalization quantity, but it will not move $L(\mathrm{H}\beta)$ perceptibly.

Table 6: Spread among the three table-interpolation schemes: the largest of the three $j(\mathrm{H}\beta)$ values divided by the smallest, in percent, on the Storey & Hummer (1995) case-B table. Rows at 5 000, 10 000 and 20 000 K and the column at $n_\mathrm{e} = 10^2\,\mathrm{cm}^{-3}$ are grid nodes, where all three schemes return the tabulated number and the spread vanishes by construction. 7024 and 8211 K were the converged benchmark temperatures of 2026-07-26 (6942 and 8176 K in the converged run `m13_restoration` of 2026-10-11).

| | $n_\mathrm{e}$ / cm$^{-3}$ | | | |
|---|---|---|---|---|
| $T_\mathrm{e}$ / K | $10^2$ | $3\times10^3$ | $10^5$ | $3\times10^6$ |
| 5 000 | 0.000 | 0.043 | 0.000 | 0.279 |
| 7 024 | 0.114 | 0.105 | 0.092 | 0.205 |
| 8 211 | 0.100 | 0.096 | 0.082 | 0.172 |
| 10 000 | 0.000 | 0.024 | 0.000 | 0.145 |
| 13 693 | 0.042 | 0.040 | 0.047 | 0.112 |
| 20 000 | 0.000 | 0.006 | 0.000 | 0.074 |

**What a solved atom says about the tables.** Having one code in the comparison that does not read the tables makes it possible to ask how good the tables are, which is otherwise unanswerable from inside this family. Two measurements were made with CLOUDY c25.00 for this memo, both single-zone, constant-temperature, hydrogen-only models with 30 resolved and 70 collapsed levels, run in the configuration of its own case-B reference test (`case b hummer no photoionization no pdest`, `no scattering escape`, `no induced processes`).

First, at the table's own grid point, $T_\mathrm{e} = 10^4$ K and $n_\mathrm{e} = 10^2\,\mathrm{cm}^{-3}$, the solved populations sit slightly *below* the tabulated case-B values, and by an amount that grows along the series:

| | H$\beta$ | H$\alpha$/H$\beta$ | Pa$\alpha$/H$\beta$ |
|---|---|---|---|
| Storey & Hummer (1995) | $1.2350\times10^{-25}$ | 2.8640 | 0.3387 |
| CLOUDY c25.00 | $1.2321\times10^{-25}$ | 2.8452 | 0.3319 |
| difference | –0.23% | –0.65% | –2.00% |

This is the quantitative version of the statement its test deck makes in words: the deviation is attributed there to the principal quantum number specification, 30/70 in the run against 100/1 in the table. The practical reading for a

table-based code is reassuring in the abstract and cautionary in the detail — the normalization line agrees to a quarter of a percent, while a ratio between widely separated series does not, at the level a Balmer/Paschen decrement measurement would notice.

Second, and this is what settles the question the table cannot answer, the same model run down to $n_e = 10^0\,\mathrm{cm}^{-3}$ shows the hydrogen coefficients essentially flat, which is the measurement quoted in the density paragraph above. Its own validation suite tests $\log n_e = 2, 4, 5, 8, 10$ and 13; the lowest is the Storey–Hummer floor, because below it there is nothing tabulated to validate against. A code that solves the populations does not need the floor, and a code that reads the table needs a defensible rule at it.

**Case A and case B.** This is where the six differ most, and where the ordering is not the one the codes' ages would suggest. MAPPINGS V 5.2.1 has the most physical treatment of any code here: the case parameter is not a switch but a *continuous* quantity computed zone by zone from the Lyman-$\gamma$ escape probability, casab = $1 - e^{-0.8597\,x}$ with $x$ the line-transfer path multiplier (`functions.f`:21–71), and the code interpolates logarithmically between the case-A and case-B tables at that value. The same escape quantity that sets the parameter also attenuates the lines in the diffuse field, so the treatment is self-consistent. M$^3$ inherits the routine and the data but not the logic: its three-dimensional driver hardcodes `caseab(1)=1.0d0` with the escape-probability call commented out one line below (`calpdf.f`:45–46, 54–55), so the connected path is pure case B everywhere and the Lyman series above Ly$\alpha$ is identically zero even in optically thin gas. MOCHII sits between the two: two explicit parameters, `par%recombination_continuum` for the ground continua (default `'transported'` since 2026-09-26, case A with the diffuse field transported; `'on_the_spot'` is case B) and `par%lyman_lines` for the line tables (default `'on_the_spot'`, case B), with the second kept separate precisely because tying them together silently loaded case-A line tables whenever the diffuse field was switched on.

The other three each conflate the two questions in a different way. IONIZE has it right by convention and says nothing about it: its lines are case B (hardcoded, no switch) while its continuum uses the case-A total recombination coefficient with an explicitly transported diffuse field — the same pairing MOCHII arrived at, with no parameter and no comment marking the distinction. CMACIONIZE hardwires case-B line coefficients while the continuum case follows from two uncoordinated defaults, a diffuse handler that is off and a recombination coefficient that is the case-A total. CLOUDY is in a category of its own here, and it is the cleanest of the seven: the case is neither a table nor a flag on the data but a *transfer condition*. `case b` sets the minimum Lyman-line optical depth to $10^5$ and case A or C to $10^{-5}$ (`parse_caseb.cpp`:19–41), and `rt_tau_reset.cpp`:137–144 then clobbers the Ly$\alpha$ optical depth to that value while every other line is transferred normally. The command also carries three options whose only purpose is to *emulate* the tabulated assumptions — `hummer` ("no collisions from n=1, 2"), `no photoionization` and `no pdest` — and its own case-B reference test uses all three, plus `no scattering escape` and `no induced processes`, with 30 resolved and 70 collapsed levels. The test's closing comment is the honest summary of what a case-B table is: "This is close to the best and most complete model of H I that the code can do. The predicted results still deviate from the tabulated 'Ca B' results because of the principal quantum number specifications (30/70 here; 100/1 in table)." In other words CLOUDY has to switch physics off to approach the tables, and the same test finds departure coefficients of 0.46–1.79 for levels $n = 30$ to 530 at $n_e = 10^2\,\mathrm{cm}^{-3}$, which is why the solved populations are the more general statement.

TORUS exposes a `caseB` flag that changes neither the lines nor the transport but replaces the hydrogen recombination

coefficient with a temperature-independent constant $2.7 \times 10^{-13}\,\mathrm{cm^3\,s^{-1}}$, and warns rather than refuses when it is combined with a transported diffuse field. MOCASSIN offers case B only.

**Density dependence, and what happens below the table.** MOCHII, MAPPINGS V 5.2.1, M$^3$ and MOCASSIN's output ratios carry it; IONIZE, CMACIONIZE, TORUS and MOCASSIN's iteration path do not; CLOUDY carries it inherently, since the populations are solved rather than read. That difference has a consequence the table-based codes have to answer for: the Storey–Hummer grids begin at $n_\mathrm{e} = 10^2\,\mathrm{cm^{-3}}$ (the original CDS files as well as the repackaged ones), so every code that reads them needs a rule below that density, and MOCHII's rule is to clamp (`sh95_mod.f90`:162). Two measurements bound what the clamp costs. From the bottom slope of the table itself, extrapolating the collisional term linearly in $n_\mathrm{e}$, the $n_\mathrm{e} \to 0$ error is $\leq 0.08\%$ for every H I and He II line in it. Independently, CLOUDY — which has no floor — gives +0.065% for H$\beta$, +0.108% for H$\alpha$ and +0.03 to +0.05% for the members up to $n = 17 \to 2$ between $n_\mathrm{e} = 10^2$ and $10^0\,\mathrm{cm^{-3}}$ at $10^4$ K. The two agree that for hydrogen the low-density limit is already reached at the table's first column, and the cost of the clamp on an integrated luminosity is smaller still: replacing it with the $n_\mathrm{e} \to 0$ limit moves $L$(H$\beta$) by −0.0002% at HII40, −0.0003% at HII20 and −0.031% in the low-density circumgalactic demonstration where every emitting cell is below the floor.

Table 7: Hydrogen recombination lines, one row per code. "Case mechanism" is how the code decides between case A and case B, which is the axis on which no two of them agree.

| Code | Emissivity source | Case mechanism | $n_\mathrm{e}$ depend. | H collis. term | Transported |
|---|---|---|---|---|---|
| MOCHII | SH95 case-A/B tables, bilinear in $(\log T_\mathrm{e}, \log n_\mathrm{e})$ | `lyman_lines`, separate from the continuum `recombination_continuum` | yes | optional, 25-level atom | no |
| MAPPINGS V 5.2.1 | SH95 absolute H$\beta$ + 90 ratios, spline in $\log T_\mathrm{e}$ | continuous, from the Ly$\gamma$ escape probability | yes | yes, with case-dependent cascade | yes, outward-only with escape factors |
| M$^3$ | the same routine and data | hardcoded case B in the 3D driver | yes | yes, as above | 5–200 eV band only, so Lyman lines only |
| MOCASSIN | ratios frozen at ($10^4$ K, $10^2$) in the iteration; SH95 interpolated at output | case B only, no switch | output ratios only | no | energy removed, photon not propagated |
| IONIZE | analytic fits to SH95, Eq. (5) | case B hardcoded, continuum case A | no | no | no |

*Table 7 continued.*

| Code | Emissivity source | Case mechanism | $n_e$ depend. | H collis. term | Transported |
|---|---|---|---|---|---|
| CMACIONIZE | the same coefficients | case B hardwired; continuum case from two uncoordinated defaults | no | no | no |
| TORUS | ratios frozen at ($10^4$ K, $10^2$) times a fitted H$\beta$ | `caseB` changes only the recombination coefficient | no | no | yes, into the diffuse redraw spectrum |
| CLOUDY | none: solved H-like level populations, $n \leq 100$ by default | a Lyman-line optical depth, $10^5$ against $10^{-5}$ | inherent | inherent | 1D line transfer |

*Helium is the exception, and it is not a small one.* The Porter et al. (2012/2013) He I table starts one decade lower, $n_e = 10^1$ cm$^{-3}$, and its 10833 Å line does not flatten there: across the lowest decade of its own grid the coefficient moves +19.1%, against $\leq 0.8\%$ for the five singlet lines in the same table. CLOUDY confirms the effect independently, +10.5% from $n_e = 10^1$ to $10^2$ cm$^{-3}$ and a further +1.1% from $10^0$ to $10^1$, against $\leq 0.04\%$ for 5876 Å. The reason is structural rather than numerical: the lower level of 10830 is the metastable $2^3$S, whose population is set collisionally, so this line has no density-independent limit in the regime the table covers — which is precisely why the $2^3$S population is computed explicitly elsewhere in MOCHII rather than inferred from a case-B emissivity. Clamping it is therefore a real error in diffuse gas, at the ten-percent level for $n_e < 10\,\mathrm{cm}^{-3}$, and the honest statement is that the tabulated 10833 emissivity should not be used below the Porter floor at all. The cost is small in an H II region and not small in a planetary nebula: from the same table, $j$(H$\beta$) rises 0.4% from $n_e = 10^2$ to $10^4$ cm$^{-3}$ at $10^4$ K and 1.6% to $10^6$, while H$\alpha$/H$\beta$ falls 2.0% by $10^6$ and 5.7% by $10^8$. This is the same axis as the metal-line one of Section 8.5, and two codes are internally inconsistent across it: CMACIONIZE and IONIZE both solve their metal lines from level populations at the local $n_e$ while leaving the hydrogen lines density-independent.

**Collisional excitation.** MAPPINGS V 5.2.1 and M$^3$ are the only codes that add a collisional term inside the line calculation itself, and they do it carefully: the excitation rate to each level (Anderson et al. 2000 collision strengths for $n \leq 5$, Giovanardi et al. fits for $6 \leq n \leq 16$) times case-dependent cascade branching fractions read from their own data file, added to the recombination brightness (`hydro.f`:756–812), with the same energy booked as cooling and not double counted. MOCHII has the equivalent as an option (`par%h_coll_effects`: a 25-level H I atom solved collisionally, its Balmer output written as separate rows so the recombination row stays a pure recombination quantity). CLOUDY needs no such term as an addition: collisional excitation is one of the rates in the level solve, which is what makes its Balmer emission depart from case B at all. IONIZE, CMACIONIZE, MOCASSIN and TORUS add nothing — consistent with the fact, recorded in the cooling audit, that none of them carries H I collisional-excitation cooling either.

**Transport.** The lines are diagnostics in MOCHII, IONIZE and CMACIONIZE: nothing re-emits a Balmer photon, and in CMACIONIZE the in-run emissivity call is commented out altogether, leaving a separate post-processing mode as the only route. MOCASSIN occupies a middle position that is easy to misread: the recombination-line energy is summed into the diffuse budget so that it correctly leaves the ionizing field, but a packet that lands in the line channel is given $\nu = 0$ and returns without being propagated — the energy is removed, the photon is not transported. TORUS genuinely transports them, writing the lines into the diffuse redraw spectrum so an absorbed ionizing packet can re-emerge as an H$\alpha$ photon. MAPPINGS V 5.2.1 transports all of them into its outward-only diffuse field with escape and dust-retention factors. M$^3$ transports the subset that falls in its hardcoded 5–200 eV Monte Carlo band, which admits the Lyman series and excludes every Balmer and redder line, and it applies no escape factor to them.

**What each code refuses to compute.** IONIZE and CMACIONIZE return zero emissivity unless $x_{\rm HI} < 0.2$ and $T_{\rm e} > 3000$ K, so their emissivity maps are blank across the ionization front and through the neutral gas by construction; CMACIONIZE's Python twin uses 0.25 where the compiled code uses 0.2. MOCHII warns once when a cell exceeds the 30 000 K table edge and clamps rather than extrapolating. MAPPINGS V 5.2.1 zeroes all hydrogen lines outside $10 < T_{\rm e} < 10^6$ K and extrapolates its ratio tables above $3 \times 10^4$ K, where they end.

**Helium, for contrast.** The same audit on helium separates the codes further. MOCHII carries Storey–Hummer He II tables in both cases and Porter et al. (2012/2013) He I emissivities; MAPPINGS V 5.2.1 and M$^3$ carry the same He II calculation and Porter et al. (2007), an earlier release from the same group; MOCASSIN uses a case-A He II table for the transported channel, which is the correct use of a case-A table for photons that ionize hydrogen. TORUS loads three helium tables at startup — Storey–Hummer He II, Benjamin et al. (1999) He I, and a Storey–Hummer H I table it never uses — and the routine that would consume them is commented out in its entirety along with both of its call sites, so TORUS emits no helium recombination line of any kind.

**Two small findings, recorded for completeness.** MOCASSIN adds its Lyman-$\alpha$ luminosity onto an existing slot of the H I array rather than a spare one, in both of its paths, and in the output path that slot reaches the user: the shipped HII40 result reports a row at $24.93\,L(\mathrm{H}\beta)$ carrying the $30 \to 8$ wavelength of 62 802 Å, which is Lyman $\alpha$ under a mid-infrared label. And in both MAPPINGS V 5.2.1 and M$^3$ the He II collisional add-on is evaluated with hydrogen's nuclear charge (`atom=2` followed by `nz=mapz(1)`, `hydro.f`:789–791), so its transition energy carries $Z^2 = 1$ instead of 4. The practical effect is negligible, since $He^+$ ground-state excitation is Boltzmann-suppressed at nebular temperatures, but it would matter in coronal gas.

## 10.3 Images and observing geometry

MOCHII writes leaf state and line-emissivity blocks to HDF5/FITS. Its Python reader deposits each AMR leaf with exact pixel-area overlap, so surface-brightness integration recovers the three-dimensional luminosity to machine precision. It also supports slab cuts and EM-, electron-, hydrogen-, or volume-weighted field maps.

For the transported FUV/EUV field, an extra converged-state pass applies observer peel-off. Stellar and diffuse emission contribute direct $e^{-\tau}$ terms, every dust interaction contributes an HG-weighted scattered term, and the optical-depth

walk uses the same gas and dust opacity as the rate calculation. Multiple observers, unattenuated direct images, and bin-resolved cubes are available. The recorded direct thin-medium gate matches the analytic image-integrated flux by +0.15% (exact hydrogenic cross sections; the gate was not rerun for this memo).

M$^3$ outputs cell-resolved physical quantities and line emissivities, and its published applications construct spatial line maps from them. This is adequate for optically thin line projection. The public driver does not show a comparable general next-event estimator for arbitrary observers, nor the separation of direct and dust-scattered continua. Thus M$^3$ is richer in the number of intrinsic lines, while MOCHII is richer in the chain that turns a three-dimensional state into calibrated synthetic observations.

# 11. Convergence, verification, and maturity

## 11.1 Convergence criteria

M$^3$ updates the radiation field, solves each illuminated cell, and counts cells whose temperature and $H^+$ changes meet the selected tolerances. The run stops when a requested fraction of ionized cells has converged or after the iteration limit. Its public defaults use percent- level cell changes and increase the packet count when convergence stalls. This is intuitive, but the result is sensitive to how the ionized-cell mask and required fraction are chosen.

MOCHII reports both cell-max and volume-integrated pairs:

$$\delta_x^{(V)} = \frac{\sum V\,|\Delta x_{\rm HII}|}{\sum V x_{\rm HII}}, \qquad \delta_T^{(V)} = \frac{\sum V n_{\rm e}\,|\Delta T_{\rm e}|}{\sum V n_{\rm e} T_{\rm e}}. \tag{7}$$

The volume criterion suppresses front-cell jitter by the small volume of the front and excludes temperature oscillations in nearly electron-free cells. In the FUV dusty test, cell maxima remained at 0.1 and 6.4 while the volume measures reached stable Monte Carlo floors of $2.7 \times 10^{-3}$ and $6.6 \times 10^{-3}$. Thirty further iterations changed the ionized volume by only 0.014%. This is a stronger convergence diagnostic for turbulent, clumpy grids, provided the tolerance is set just above the packet-noise floor. Since 2026-09-26 the volume test (`par%conv_crit = 'vol'`) no longer relies on that choice alone (`src/fixed_point_distance_mod.f90`): the decay rate $\rho$ is fitted over the last 8 iterations and, while the step still decays ($\rho < 0.95$), the remaining distance $d_n\rho/(1-\rho)$ must fall below the tolerance; once it no longer decays, the step must be below the tolerance *and* the signed change of the ionized volume must show no mean drift resolvable by a two-sided 95% Student $t$ test over the same 8 iterations. An earlier test on the step alone had stopped the HII40 benchmark at 37 iterations with the He I front 0.17 pc short.

## 11.2 Validation evidence

The MAPPINGS lineage has the stronger external maturity record. M$^3$ inherits decades of MAPPINGS development and has been published against the HII20, HII40, and PN150 Lexington/Meudon standards. Its three-dimensional blister, bipolar, and fractal applications demonstrate the importance of geometry for spatially resolved line diagnostics [1, 2]. Because both codes have now been published against HII20 and HII40, the two line lists can be set side by side without running anything; that is Section 11.3, and it is the one place in this memo where a MOCHII number is compared with an M$^3$ number on the same input.

MoCHII is newer, but its current repository contains unusually granular gates:

- analytic H/He attenuation and rate integrals;
- a Strömgren sphere against an exact one-dimensional reference;
- AMR versus uniform and solution-driven refinement versus native high resolution;
- H/He thermal balance against an independent Gauss–Seidel reference;
- direct CHIANTI fit evaluation and PyNeb line-ratio checks;
- MOCASSIN HII20/HII40 state and line comparisons;
- case-A/case-B diffuse-field bracketing;
- dust absorption/scattering brackets and IR energy closure;
- direct peel-off flux against an analytic observer solution; and
- exact three-dimensional emissivity-to-map luminosity conservation.

These gates provide strong internal evidence and make regressions easy to localize. They do not yet replace the independent user base and long publication history of MAPPINGS. The correct conclusion is that the MAPPINGS lineage is more mature externally, while MoCHII is more explicit and fine-grained in its current numerical audit trail. MAPPINGS V 5.2.1 does not have a Monte Carlo convergence problem: zone step control and the global shock/precursor iteration dominate instead. Its decades of use and broad model suite are a major maturity advantage, but the inspected 5.2.1 tree does not provide a gate hierarchy as explicit as the current MoCHII test directories. A controlled local benchmark is still required before attributing any line difference to geometry rather than to the 5.1.13-versus-5.2.1 atomic and solver divergence.

## 11.3 M$^3$'s published benchmark fluxes against MoCHII's

Both codes have now been published against the same standards, so the comparison can be made from the two papers without running either. Jin et al. (2022) Tables 2 and 3 give M$^3$ at $33^3$ and $55^3$ cells beside MAPPINGS, CLOUDY, MOCASSIN3D, RR and PH, with the median and $1\sigma$ spread of those five [1]. MoCHII's column is `results/continuous_energy/m11_hecx_solver/hii{40,20}_converged` (2026-09-27), the production inputs under the defaults of that date, run to the converged stopping test (m11; not rerun under the 2026-10-11 physics: every MoCHII line ratio of this subsection is from that run, before the restoration; only the H$\beta$ rows carry the converged `m13_restoration` luminosity, which moved +0.10% and +0.02%), with the blends summed as in the paper's table script (an earlier version of this section read `m1_oxfix` of 2026-07-31, a run that the old stopping test had ended early); its composition matches Jin Table 1 entry for entry (He 0.1, C $2.2\times10^{-4}$, N $4.0\times10^{-5}$, O $3.3\times10^{-4}$, Ne $5.0\times10^{-5}$, S $9.0\times10^{-6}$), so nothing has to be corrected for abundance. The M$^3$ column below is the $55^3$ one.

Table 8: HII40. Line fluxes relative to H$\beta$, except H$\beta$ itself. "Med$\pm\sigma$" is Jin et al.'s median and spread over MAPPINGS, CLOUDY, MOCASSIN3D, RR and PH; it is *not* the ten-column reference set used elsewhere in this memo.

| Line | M$^3$ | Med | $\sigma$ | MoCHII | $\frac{\text{MoCHII}}{\text{M}^3}$ |
|---|---|---|---|---|---|
| H$\beta$ [$10^{37}$ erg s$^{-1}$] | 1.98 | 2.05 | 0.03 | 2.025 | 1.023 |
| He I 5876 | 0.115 | 0.116 | 0.047 | 0.1120 | 0.974 |
| C II] 2325+ | 0.164 | 0.164 | 0.027 | 0.1761 | 1.074 |
| C III] 1907+1909 | 0.076 | 0.071 | 0.013 | 0.0556 | 0.732 |
| [N II] 122 $\mu$m | 0.029 | 0.030 | 0.003 | 0.0319 | 1.099 |
| [N II] 6584+6548 | 0.632 | 0.723 | 0.082 | 0.6908 | 1.093 |
| [N II] 5755 | 0.0053 | 0.0053 | 0.0007 | 0.00588 | 1.109 |
| [N III] 57.3 $\mu$m | 0.297 | 0.294 | 0.029 | 0.2882 | 0.970 |
| [O I] 6300+6363 | 0.0290 | 0.0094 | 0.0150 | 0.02812 | 0.970 |
| [O II] 7320+7330 | 0.027 | 0.027 | 0.005 | 0.03433 | 1.271 |
| [O II] 3726+3729 | 2.05 | 2.05 | 0.14 | 2.261 | 1.103 |
| [O III] 52+88 $\mu$m | 2.55 | 2.34 | 0.11 | 2.446 | 0.959 |
| [O III] 5007+4959 | 2.37 | 2.17 | 0.26 | 2.364 | 0.997 |
| [O III] 4363 | 0.00421 | 0.0035 | 0.0008 | 0.00437 | 1.039 |
| [Ne II] 12.8 $\mu$m | 0.199 | 0.199 | 0.015 | 0.2062 | 1.036 |
| [Ne III] 15.5 $\mu$m | 0.290 | 0.294 | 0.066 | 0.2818 | 0.972 |
| [Ne III] 3869+3968 | 0.069 | 0.083 | 0.013 | 0.0703 | 1.018 |
| [S II] 6716+6731 | 0.194 | 0.141 | 0.045 | 0.3599 | 1.855 |
| [S II] 4068+4076 | 0.0086 | 0.0070 | 0.0021 | 0.01547 | 1.799 |
| [S III] 18.7 $\mu$m | 0.534 | 0.567 | 0.033 | 0.5021 | 0.940 |
| [S III] 9532+9069 | 1.04 | 1.21 | 0.10 | 0.9876 | 0.950 |

At HII40 the two codes agree within 10% on 15 of the 21 entries and the median ratio is 1.02. More usefully, they leave the 1995-era median in the same direction wherever either of them leaves it: H$\beta$ low (0.97, 0.99 of the median), [O I] 6300+6363 far high (3.1$\times$ the median for M$^3$, 3.0$\times$ for MoCHII), [O III] 4363 high (1.20, 1.25), [O III] 52+88 $\mu$m high (1.09, 1.05), both [S II] doublets high, both [S III] diagnostics low, [Ne III] 3869+3968 low. The six entries outside 10% are C III] 1907+1909 (0.73), [N II] 5755 (1.11), [O II] 7320+7330 (1.27), [O II] 3726+3729 (1.10), and the two [S II] doublets (1.86, 1.80); [O I] 6300+6363, outside at 0.83 on the early-stopped run, agrees to 0.97 on the converged one. That agreement is weaker evidence than it looks, and Jin et al. supply the reason themselves. Their Section 8 measures the resolution sensitivity of each line between $33^3$ and $55^3$: most move under a few percent, but [O I] 6300+6363 moves 32%, [O II] 7320+7330 moves 35%, [S II] 6716+6731 moves 23% and [S II] 4068+4076 moves 20%. Those are exactly the lines on which the two Monte Carlo codes exceed the median together, and they are the lines that form in the ionization front, which both codes resolve on a Cartesian mesh where the one-dimensional codes integrate it adaptively. Two codes agreeing because they share a method is not independent confirmation of the physics — the same trap that cost a retracted conclusion in `md/HII40_CLOUDY_RESIDUAL_INVESTIGATION.md`.

Two entries are informative in the other direction. He I 5876 was the one line where M$^3$ sat on the median (0.115 against 0.116) and the early-stopped MOCHII run did not (0.1067, 7% low, at HII40 and not at HII20). Converged, it is 0.1120, 0.97 of M$^3$ and inside the published 0.112–0.119. The deficit was the premature stop that also held back the HII40 helium front, not a helium-physics difference between the codes: on the same physics, `m4_nitrogen` gives 0.1051 where the old test stopped it (37 iterations) and 0.1130 after 60. And [O III] 5007+4959, 4363 and 52+88 $\mu$m all agree with M$^3$ to 0.3–4% while sitting 4.5–25% above the median, which says the oxygen result is a two-code statement rather than a MOCHII one. Under the restored defaults the 5007/4959 split moves for a physical reason: the CHIANTI O III atom of the default model carries the Storey & Zeippen (2000, MNRAS 312, 813) Table 3 $A$ values, $6.791 \times 10^{-3}$ and $2.046 \times 10^{-2}$ s$^{-1}$, with the relativistic corrections to the M1 operator, and the two lines share their upper level, so their ratio is $A$ times photon energy alone: 2.984 in `m13_restoration`, where CHIANTI's own `o_3.wgfa` and CLOUDY c25.00 give the Galavis et al. (1997) 2.89. At HII40, 5007 rises 2.0% and 4959 falls 1.2% against `m11`; a 5007 difference against CLOUDY has to be read net of that.

**The sulfur result, and where it comes from**

HII20 separates MOCHII from M$^3$ where HII40 did not.

Table 9: HII20, columns as in Table 8.

| Line | M$^3$ | Med | $\sigma$ | MOCHII | $\frac{\text{MoCHII}}{\text{M}^3}$ |
|---|---|---|---|---|---|
| H$\beta$ [$10^{36}$ erg s$^{-1}$] | 4.81 | 4.89 | 0.09 | 4.847 | 1.008 |
| He I 5876 | 0.0071 | 0.0072 | 0.0016 | 0.00702 | 0.989 |
| C II] 2325+ | 0.059 | 0.059 | 0.010 | 0.0732 | 1.241 |
| [N II] 122 $\mu$m | 0.072 | 0.071 | 0.001 | 0.0729 | 1.012 |
| [N II] 6584+6548 | 0.783 | 0.803 | 0.056 | 0.8473 | 1.082 |
| [N II] 5755 | 0.0028 | 0.0028 | 0.0003 | 0.00330 | 1.177 |
| [N III] 57.3 $\mu$m | 0.0027 | 0.0027 | 0.0007 | 0.00391 | 1.448 |
| [O I] 6300+6363 | 0.0241 | 0.0088 | 0.0091 | 0.01828 | 0.759 |
| [O II] 7320+7330 | 0.0064 | 0.0080 | 0.0017 | 0.01252 | 1.957 |
| [O II] 3726+3729 | 1.24 | 1.17 | 0.13 | 1.427 | 1.151 |
| [O III] 52+88 $\mu$m | 0.0029 | 0.0029 | 0.0007 | 0.00341 | 1.177 |
| [O III] 5007+4959 | 0.0019 | 0.0014 | 0.0005 | 0.00218 | 1.145 |
| [Ne II] 12.8 $\mu$m | 0.296 | 0.290 | 0.013 | 0.3030 | 1.024 |
| [S II] 6716+6731 | 0.529 | 0.499 | 0.039 | 0.806 | 1.524 |
| [S II] 4068+4076 | 0.016 | 0.015 | 0.003 | 0.02388 | 1.493 |
| [S III] 18.7 $\mu$m | 0.324 | 0.371 | 0.042 | 0.2120 | 0.654 |
| [S III] 9532+9069 | 0.441 | 0.526 | 0.060 | 0.2987 | 0.677 |

MOCHII's two S$^+$ diagnostics are high by 1.62 and 1.59 against the median and its two S$^{2+}$ diagnostics low by 0.571 and 0.568. Each pair moves by one factor, and each pair contains one infrared fine-structure line whose emissivity barely depends on $T_e$ and one temperature-sensitive optical line: a $T_e$ error or a level-population error would separate the

members of a pair, and a shift in the ion fraction would not. Weighted by emission measure ($n_e n_p\, dV$), MOCHII's HII20 sulfur is $S^+$ 0.613 / $S^{2+}$ 0.387 / $S^{3+}$ 0.000; carrying both pairs onto the median needs about 0.39 / 0.62, that is an $S^+/S^{2+}$ ratio too high by a factor near 2.5. HII40 carries the same sign ($S^+$ 0.133 / $S^{2+}$ 0.822 / $S^{3+}$ 0.045; [S II] 2.55 and 2.21 times the median, [S III] 0.89 and 0.82).

The cause is in the atomic data, and both trees can be read directly.

*The photoionization cross section is excluded.* MOCHII's $S^+$ outer-shell fit (`data/atomic/element_s.txt`, `TRANSITION 2`: $E_0 = 8.787$, $\sigma_0 = 313.6$ Mb, $y_a = 3.442$, $P = 12.81$, $y_w = 0.7354$, $y_0 = 2.782$, $y_1 = 0.1788$) and M$^3$'s (`lab/data/ionisation/PHIONDAT2.txt`:82) are the same seven numbers; MOCHII's first $S^+$ inner-shell edge is at 184.6 eV (`PHINNER`), so in the band that fit is its whole cross section. Whatever separates the codes on sulfur, it is not the cross section.

*The recombination is not.* MOCHII takes $\alpha(S^{2+} \to S^+)$ from Badnell, radiative plus dielectronic; at 6926 K the dielectronic term is 2.2 times the radiative one. M$^3$ reads its radiative term from `RRECOMDAT.txt`:154 as the power law $1.8\times10^{-12}(T/10^4)^{-0.686}$, within 10% of MOCHII's, and its dielectronic term from `DRECOMDAT.txt`:163, a Shull & Van Steenberg (1982) two-term fit whose lowest excitation energy is $8.2\times10^4$ K [8] and which is therefore six orders of magnitude below the radiative term at nebular temperatures. The totals:

| $T$ [K] | MOCHII | M$^3$ | ratio |
|---|---|---|---|
| 5000 | $8.38\times10^{-12}$ | $2.90\times10^{-12}$ | 2.89 |
| 6926 | $6.68\times10^{-12}$ | $2.32\times10^{-12}$ | 2.88 |
| 8170 | $5.97\times10^{-12}$ | $2.08\times10^{-12}$ | 2.87 |
| 10000 | $5.23\times10^{-12}$ | $1.84\times10^{-12}$ | 2.85 |
| 20000 | $3.81\times10^{-12}$ | $2.06\times10^{-12}$ | 1.85 |

in cm$^3$ s$^{-1}$. Since $n(S^+)/n(S^{2+}) = n_e\alpha/\Gamma$ and the two codes share $\sigma_{pi}$, a factor 2.9 in $\alpha$ is a factor 2.9 in the ratio at fixed radiation field — the right sign and about the right size for the factor near 2.5 the line ratios ask for. The whole difference is the low-temperature dielectronic recombination of $S^{2+}$, which Badnell computed and Shull & Van Steenberg's fit does not reach.

*A third code sides with* MOCHII. COLT carries the same rate as two inline functions, `alpha_RR_SIII` (`rates.h`:1539) and `alpha_DR_SIII` (`:1612`), and every constant in both is MOCHII's: $2.478\times10^{-11}$, 0.4642, 329.4, $2.166\times10^7$, 0.3351, $7.63\times10^5$ for the radiative term and the same seven $(c_i, E_i)$ pairs for the dielectronic one. Two independent implementations reading the same Badnell tables is not an independent determination — it is the same source twice, the trap recorded in the physics lessons — but it does settle where the disagreement lives. It is M$^3$ that is alone, and the disagreement is a choice of compilation rather than an error in either code.

The direction of the modern data is worth stating before the decisive test below. MAPPINGS's own tree carries a *third* determination for this one rate, `lab/data/ionisation/RECOMDAT2.txt`, headed "Badnell 2015 empirically adjusted DR and RR recombination for SIII → SII". Summing its three tabulated levels as its own header prescribes gives 1.89 times MOCHII's Badnell total at 6926 K and 5.4 times the rate M$^3$ actually uses; how those levels should be combined is a question this memo does not settle, and the number is quoted with that caveat. What is not in doubt is that no source file in the M$^3$ tree reads it — `grep` for `RECOMDAT2` over `src/` returns nothing, while `RRECOMDAT` and `DRECOMDAT` are

opened in `mapinit.f`:1323 and `:1464` — so it is present and unconnected, which is precisely the level-1-versus-level-2 distinction of Section 1..

*And* CLOUDY *c25.00 settles it.* The check named above has now been run. CLOUDY's own test suite carries this benchmark as `tsuite/auto/hii_coolstar.in` — "dust free cool HII region model, Lexington 1995", a 20 kK blackbody at $Q(\mathrm{H}) = 10^{49}$, $n_\mathrm{H} = 100$, $R_\mathrm{in} = 3\times10^{18}$ cm, on the same abundances — and it was run here unchanged except for lowering `print line faint` so the weak lines print (c25.00, 190 zones, 2 iterations, 55 s, all 30 monitors passing). Set beside the same median:

| | Med | M$^3$ | CLOUDY c25.00 | MOCHII |
|---|---|---|---|---|
| [S II] 6716+6731 | 0.499 | 0.529 | 0.7441 | 0.8063 |
| ratio to median | | 1.06 | **1.49** | **1.62** |
| [S II] 4068+4076 | 0.015 | 0.016 | 0.0226 | 0.0239 |
| ratio to median | | 1.07 | **1.51** | **1.59** |
| [S III] 18.7 $\mu$m | 0.371 | 0.324 | 0.2090 | 0.2120 |
| ratio to median | | 0.87 | **0.56** | **0.57** |
| [S III] 9532+9069 | 0.526 | 0.441 | 0.2933 | 0.2987 |
| ratio to median | | 0.84 | **0.56** | **0.57** |

CLOUDY c25.00 reproduces MOCHII's excess and deficit on all four diagnostics to 1.4–8.4% (MOCHII/CLOUDY = 1.084, 1.058, 1.014, 1.018), and both sit at $\sim$1.5–1.6$\times$ the 1995-era median on $S^+$ and $\sim$0.57$\times$ on $S^{2+}$ while M$^3$ stays near it. Over the seventeen entries of Table 9 the two agree to a median ratio of 1.031, fourteen of them within 10% (on the early-stopped `m1_oxfix` run, 1.033 and fifteen), and the volume-weighted $T_\mathrm{e}(\mathrm{H}^+)$ agrees to 0.3% — 6923.1 K against MOCHII's 6944.7 K ($\int T_\mathrm{e} n_\mathrm{p}\, dV / \int n_\mathrm{p}\, dV$ on the m11 run; 6926 K on `m1_oxfix`). **The [S II] excess and the [S III] deficit are what modern low-temperature dielectronic recombination of $S^{2+}$ does to the sulfur balance, and the Lexington median encodes the older data. They are not a MOCHII defect.**

Two independent confirmations come with that run. The recombination files are the same bytes: `badnell_dr.dat` and `badnell_rr.dat` of CLOUDY c25.00 are md5-identical to MOCHII's, both headed `(C)20230512` and `(C)20230511`, and the $(16, 14, 1)$ rows are the constants quoted above. And CLOUDY's regression history names the mechanism and dates it. In `hii_coolstar.in` the monitor for [S II] 6720 carries `>>chng 14 dec 14, from 0.6213 to 0.7149, update S2+ - S+ DR to Badnell+ 2015 ApJ`, a +15% step, and the three [S III] monitors carry mirrored downward steps of the same date — 0.2735 $\rightarrow$ 0.2325 at 18.7 $\mu$m, 0.4458 $\rightarrow$ 0.3789 at 33.5 $\mu$m, 0.269 $\rightarrow$ 0.2317 at 9531 Å — followed by `>>chng 23 may 11, update RR DR to 2022 Badnell work`, which is the release both codes now read. So there are three vintages of one rate in play: Shull & Van Steenberg (1982), which is what M$^3$ actually runs on; the 2015 adjustment, which MAPPINGS ships unread; and the 2023 release, which MOCHII and CLOUDY both read. The two codes on the newest data agree with each other and disagree with the benchmark.

*[O I] is a different question and the same run separates it.* CLOUDY c25.00 gives [O I] 6300+6363 = 0.0084, sitting on the median 0.0088, against MOCHII's 0.0183 (0.0158 on the early-stopped run) and M$^3$'s 0.0241. No data-vintage argument covers that: it separates the two three-dimensional Monte Carlo codes from both one-dimensional references together, which is the front-resolution reading of the paragraphs above and not an atomic-data one. The same holds at a smaller size for [O II] 7320+7330, where CLOUDY (1.41$\times$ the median) sits with MOCHII (1.57) rather than with M$^3$

(0.80) — so that one is *not* front resolution, and it is unexplained.

This has the same shape as the $\alpha_A(He\,\mathrm{I})$ caution already recorded for MOCASSIN and the $\alpha_B(He\,\mathrm{I})$ one for COLT: a cross-code disagreement traced to data vintage, to be subtracted before it is read as a transport or geometry error.

## 12. Software architecture and maintainability

MOCHII uses Fortran modules, allocatable leaf arrays, a single namelist, MPI-3 shared memory, HDF5/FITS sections, and a data-driven species registry. New atomic products are generated offline by Python tools and carry machine-readable provenance. A new ion normally requires no new solver code. The transport, ion balance, thermal balance, lines, dust, and output layers are separated modules. This architecture substantially reduces the cost and risk of adding physics.

The public M$^3$ tree retains the Fortran 77 MAPPINGS architecture: large common blocks, compile-time maxima, interactive input, and many global arrays. Several historical `montph7_*` variants duplicate large parts of the transport driver. This structure has the advantage of preserving a mature code base and its full menu of MAPPINGS models, but it makes a change to the 3D algorithm harder to propagate and test. The available Sphinx input/output documentation is sparse, so reliable use often requires reading the source and reproducing the authors' setup.

The architectural verdict is therefore clear: MOCHII is easier to extend, audit, automate, and connect to simulation data. M$^3$'s value lies primarily in the depth of the inherited physical library, not in modern software structure.

MAPPINGS V 5.2.1 shares the same common-block and interactive heritage, but its organization is cleaner than the collection of duplicated M$^3$ `montph7_*` experiments and it has a single public `map52` model selector. Runtime preference files make the 16- or 30-element atom sets configurable without recompilation. Nevertheless, automated model grids require scripting interactive input and parsing many ASCII outputs; the current documentation overview explicitly remains incomplete. For maintainability the ordering is MOCHII first, MAPPINGS V 5.2.1 second, and the experimental M$^3$ 3D fork third.

## 13. Suitability by science case

Table 10: Recommended code by present science requirement.

| Science requirement | Preferred code | Reason |
|---|---|---|
| Clumpy or porous H II region from a TNG/RAMSES snapshot | MOCHII | AMR, front refinement, and simulation-oriented I/O. |
| Ionization-front position, shadows, and clump skins | MOCHII | Adaptive resolution and the analytic absorption estimator. |
| Dusty Strömgren structure and UV attenuation | MOCHII | Gas–dust competition and validated HG scattering in the 3D band. |

*Table 10 continued.*

| Science requirement | Preferred code | Reason |
|---|---|---|
| PAH rings, IR SEDs, or 8–160-$\mu$m maps | MOCHII | SEDust, leaf dust emissivities, and exact absorbed-energy normalization. |
| Spatially resolved IFU line maps | Usually MOCHII | AMR emissivities and flux-conserving projection; verify that the required ions exist. |
| Very broad integrated optical/UV/IR line inventory | MAPPINGS V 5.2.1 | Current full 30-element MAPPINGS preference set and deterministic detailed spectrum. |
| Compact high-density nebula | MAPPINGS V 5.2.1 if 1D; M$^3$ if 3D | Both feed density-dependent multilevel cooling back into $T_e$; geometry decides. |
| Hard PN-like ionizing spectrum | MAPPINGS V 5.2.1 if 1D; M$^3$ if 3D | More high charge states; M$^3$ additionally has published PN150 3D validation (MOCHII carries stages to VII since 2026-10-11, but its PN150 run has not converged yet). |
| Non-Maxwellian $\kappa$ electron experiment | MAPPINGS V 5.2.1 or M$^3$ | Both MAPPINGS-derived trees have native $\kappa$ rate support. |
| One-dimensional radiation-pressure-confined nebula | MAPPINGS V 5.2.1 | Isobaric photoionization can include the accumulated radiation pressure. |
| Finite source lifetime or source-off ionization history in 1D | MAPPINGS V 5.2.1 | P6/P7 connect finite-age modes to the time-dependent ion solver. |
| Steady 1D radiative MHD shock and precursor | MAPPINGS V 5.2.1 | S5 couples adaptive post-shock cooling to an iterated auto-preionization solution. |
| CIE, NEQ, or PIE cooling-curve grid | MAPPINGS V 5.2.1 | Dedicated CC, NC, and PP drivers and a broad general-plasma ion inventory. |
| Controlled test of modern CHIANTI v11 coefficients | MOCHII | Versioned fitted data and direct fit-error gates. |

*Table 10 continued.*

| Science requirement | Preferred code | Reason |
|---|---|---|
| Three-dimensional time-dependent ionization | None | Both public 3D drivers are equilibrium solvers; MAPPINGS V 5.2.1 time dependence is 1D. |
| Three-dimensional radiative shock | None | MAPPINGS V 5.2.1 has a strong 1D shock mode, but it is not coupled to either 3D transport architecture. |
| Molecular $H_2$/CO PDR | None | All three need a dedicated molecular chemistry and line-transfer layer. |
| Ly$\alpha$ or metal resonance-line profiles | None | Requires frequency-redistributing resonant transfer, e.g. LaRT. |

# 14. Recommended development path for MoCHII

The comparison suggests a focused roadmap rather than wholesale replacement of either code.

1. **Put density dependence into the thermal cooling.** Done 2026-07-26 (`par%cooling_model = 'local_ne'`, the default): the generic $n$-level solver builds a $\Lambda(T_e, n_e)$ suppression table for every ion at setup (`src/nlevel_cooling_mod.f90`), which the thermal loop reads. It closed the most consequential gap relative to MAPPINGS; an in-loop solve for every ion remains the MAPPINGS arrangement.

2. **Extend the registry upward in ion stage.** Add the ions needed for PN and hard-field applications before adding marginal new low stages. Done 2026-10-11 to stage VII for all eleven metals, as a data operation (`tools/fitting/make_element_data.py`); the PN150 gate on it is pending, and stages beyond VII and the Auger stage coupling remain open.

3. **Add Na, Al, and Ni only when driven by a diagnostic need.** Their MAPPINGS implementations provide useful reference behavior, while the MOCHII data files should retain explicit provenance and ion-by-ion gates.

4. **Retain the present transport architecture.** The octree, analytic estimator, Cartesian DDA, continuous photon-energy sampling, and peel-off layer are major advantages and should not be replaced by the legacy M$^3$ packet synchronization model.

5. **Use MAPPINGS V 5.2.1 as the differential microphysics oracle.** Construct one-zone tests at fixed $(T_e, n_e, J_\nu)$ and compare ion fractions, heating/cooling terms, and individual line emissivities before comparing full 3D nebulae. Use M$^3$ only as the second-stage test of how that inherited microphysics behaves in arbitrary geometry.

6. **Perform a controlled common-physics benchmark.** Use the same uniform grid, source SED, abundances, dust-off setting, frequency range, and atomic subset. Compare $J_\nu$, photo-rates, ion fractions, $T_e$, cooling budgets, and line emissivities separately. Only then compare total runtime and MPI scaling.

The most valuable hybrid would combine MAPPINGS V 5.2.1-like breadth of microphysics — more ion stages and elements; the density-dependent line cooling is already in MOCHII through its table — with the current MOCHII geometry, transport, dust, and observer layers. This would preserve the strongest feature of each code without inheriting the principal numerical and software limitations of the other.

## 15. Final assessment

| Capability | Stronger now | Confidence |
|---|---|---|
| Three-dimensional transport estimator | MOCHII | High |
| Adaptive spatial resolution | MOCHII | High |
| Dust scattering and IR/PAH observables | MOCHII | High |
| Observer imaging and data products | MOCHII | High |
| Atomic-data provenance and fit auditability | MOCHII | High |
| Element and ion-stage breadth | MAPPINGS V 5.2.1 | High |
| Density-dependent thermal line cooling | Comparable (MOCHII by table) | High |
| Hard PN/general plasma coverage | MAPPINGS V 5.2.1 | High |
| Finite-age 1D photoionization | MAPPINGS V 5.2.1 | High |
| Radiative MHD shock and precursor | MAPPINGS V 5.2.1 | High |
| Noise-free 1D parameter-grid throughput | MAPPINGS V 5.2.1 | High |
| External validation history | MAPPINGS lineage | High |
| Software extensibility and automation | MOCHII | High |
| Absolute line accuracy for every ion | Case dependent | Requires a common-data benchmark |
| Massively distributed memory ceiling | Architecture dependent | Requires a scaling experiment |

For the principal intended application–dusty, structured, spatially resolved H II regions–MOCHII is the better current platform. For broad integrated emission-line diagnostics, dense nebulae, harder ionization conditions, finite-age 1D photoionization, and radiative shocks, MAPPINGS V 5.2.1 is the more complete physical reference. M$^3$ becomes preferable only when those MAPPINGS-like line and cooling strengths must be combined with arbitrary three-dimensional Cartesian geometry. None of these judgments is universal: first decide whether geometry, time history, shock dynamics, dust observables, density-dependent cooling, or ion-stage breadth is the controlling requirement.

# 16. Hard-spectrum (PN and AGN) extension plan

This section records the staged plan that follows from the comparison (kept as `md/HARD_SPECTRUM_PLAN.md`, drafted 2026-07-18). It uses the same staged-gate discipline as `md/PLAN.md`.

*Status 2026-09-27.* Three pieces have been done since the draft, each recorded where it belongs below: the He II excited-channel diffuse photons of HS3 (the He III case-B cascade, `par%heii_diffuse`, 2026-08-09), the density-dependent cooling of HS7 (as `par%cooling_model = 'local_ne'`, 2026-07-26, in a different form from the one drafted), and the shell-resolved opacity of HS6 part 1 (2026-09-24). The stellar and diffuse photon energies are now sampled continuously, so the plan's items about the aligned-edge band builder, `nnu_ion`, and `ion_bin_of` concern the diagnostic $J_\nu$ tally only; and `launch_sequence='sobol'` has been the default since 2026-09-25. HS1, HS2, HS4, HS5, and HS6 parts 2–4 are open.

*Status 2026-10-11.* HS2 is done: stages I–VII of all eleven metals (77 stages, 66 transitions; `MAX_ION_STAGES` = 7, and $\bar{g}_{\rm ff}$ to $Z_{\rm ion}$ = 6), with the full CHIANTI 11.0.2 atoms as the default excitation model. The `phfit2` subshell sum of HS6 part 1 is the default and the only form (it had been replaced by the outer shell alone in the tree of October 9–11 and is restored). The metal ground free–bound continua and the free–free inside the band are diffuse channels 6 and 7 (the M3 item of `md/METAL_COUPLING_PLAN.md`, and the free–free item recorded in `md/HARD_SPECTRUM_PLAN.md`). Of HS1, the band is refused above the terminal ionization potential of a present element. Open: the HS4 runs (PN150 in `results/continuous_energy/m13_restoration/`, stopped unconverged and to be rerun), HS5, and HS6 parts 2–4, the Auger yields, stage matrix and Auger-electron energetics, planned as `par%auger_ionization`.

## 16.1 Scope and the two tracks

"Hard spectrum" splits into two physically distinct regimes, kept as separate tracks because they need different physics:

- **PN track** — thermal central stars, $T_{\rm eff} \sim$ 75–200 kK. A 150 kK blackbody has $kT$ = 12.9 eV; its photon flux at 300 eV sits $\sim 23\,kT$ out on the Wien tail ($e^{-23} \sim 10^{-10}$ before the phase-space factors), and the lowest K-shell edge of an abundant metal (C, 290 eV) is never reached with significant flux. The MOCASSIN PN150 benchmark itself integrates its band only to `nuMax` = 24 Ryd = 326 eV. The PN track therefore needs *no inner-shell/Auger physics and no Compton terms* — it is band headroom, higher ion stages, the He II diffuse channels, and (at PN densities) density-dependent cooling.

- **AGN track** — a power-law/big-bump continuum extending to keV X-rays. Above $\sim$0.3 keV inner-shell absorption dominates the metal opacity and Auger multi-electron ejection couples non-adjacent ion stages; near $\sim$0.5–1 keV the Thomson cross section ($6.65 \times 10^{-25}$ cm$^2$) overtakes the steeply falling ($\sigma \propto E^{-3}$) photoabsorption of the residual neutrals, and Compton heating enters the energy budget. The first increment caps the band at $\sim$500 eV, where photoabsorption still dominates, and defers the Compton terms (Section 16.10).

Current baseline (v1.00-dev, 2026-07-18): band 13.6–100 eV (+FUV option); registry stages C 4 / N 3 / O 3 / Ne 3 / S 4 / Ar 4 / Mg 3 / Fe 4 / Si 5 / Cl 5 / Ca 5, with the compile-time stage ceiling `MAX_ST` = 6 in `species_mod` — the PN targets below fit it exactly, so no framework change is needed there; `te_max` = $5 \times 10^4$ K and the Tier-1 cooling fits are generated over $10^3$–$10^5$ K; $\bar{g}_{\rm ff}$ is tabulated for $Z_{\rm ion}$ = 1–4 (`cooling_mod`, parameter `NGZ`); secondary

ionization (`use_sec_ion`, Shull–van Steenberg 1985) is implemented but default off; the Owen-scrambled Sobol launch is available. The He II Storey–Hummer lines (case A and B), the He I Porter lines, and the diffuse He III ground channel already cover the He II 4686 diagnostic side, and the aligned-edge band builder (`ion_align_edges`, default on) with its graceful threshold-dropping logic is exactly the piece that makes many new in-band edges safe.

## 16.2 Reference assets (verified present)

Table 11: Local reference assets for the hard-spectrum work.

| Asset | Use |
|---|---|
| MOCASSIN PN150 + PN75 benchmark inputs (`benchmarks/gas/{PN150,PN75}`) | primary 3D Monte Carlo gates; run MOCASSIN itself *converged* (the HII40 lesson). PN150 input read directly: 150 kK blackbody, $n_{\rm H} = 3000$ cm$^{-3}$, $R_{\rm in} = 10^{17}$, $R_{\rm out} = 4.07 \times 10^{17}$ cm, band to 24 Ryd, abundances He 0.10, C $3 \times 10^{-4}$, N $10^{-4}$, O $6 \times 10^{-4}$, Ne $1.5 \times 10^{-4}$, Mg $3 \times 10^{-5}$, Si $3 \times 10^{-5}$, S $1.5 \times 10^{-5}$ (`abunPN150.in`) — note Mg and Si ARE in the composition, which drives their stage targets below. |
| Cloudy c23.01 test suite (`tsuite/auto/pn_paris*.in`, `nlr_paris.in`, `agn_lex00_u{0,1,m1}.in`) | 1D cross-code gates: Meudon PN, AGN NLR, and the Lexington-2000 X-ray slabs. The `agn_lex00` inputs read directly: $n_{\rm H} = 10^5$ cm$^{-3}$, $\phi$(H) = $10^{15.477}$ cm$^{-2}$s$^{-1}$, stop column $10^{16}$ cm$^{-2}$, and a *tabulated* SED (Cloudy `interpolate/continue` points) — so MOCHII reproduces the illumination exactly through the existing `spectrum_type` file route, without waiting for the AGN preset. |
| Cloudy AGN SED (`source/parse_agn.cpp`) | the standard AGN continuum parameterization (big-bump $T$, $\alpha_{\rm ox}$ anchored at 2500 Å–2 keV, $\alpha_{\rm uv}$, $\alpha_{\rm x}$) for a MOCHII preset. |
| Kaastra & Mewe (1993) Auger yields (`data/mewe_nelectron.dat`, `mewe_fluor.dat`) | electrons ejected per inner-shell vacancy, shell-resolved, machine-readable provenance for the Auger stage. |
| Verner & Yakovlev (1995) shell-resolved cross sections (`sigma_vy95` already in `photo_xsec.f90`) | inner-shell photoionization above the K/L edges. |
| CHIANTI v11.0.2 + Badnell 2023 RR/DR (in-tree) | levels/collision strengths and recombination for every new ion stage (coverage verified). |

## 16.3 HS1: band headroom

Mostly validation; the code changes are one-line parameters.

1. `eion_max` (define.f90, default 100 eV) raised to 300–500 eV for hard runs. This matters even for stages the registry ALREADY tracks: the Ne IV→V threshold is 97.1 eV — just under the present cap — so today's band cannot photoionize to Ne V at all, and the new O/Ne thresholds (113.9, 126.2 eV) lie beyond it. Cross-section validity is not the issue (the Verner et al. fits extend to keV energies and H I/He II are exact hydrogenic); the work is verifying the pieces that ASSUME the band shape: the aligned-edge builder with ∼30–40 in-band thresholds once HS2 lands (recommend `nnu_ion` $\geq 64$, 128 clean; the largest-remainder allocator and the metal-edge dropping logic already degrade gracefully), the FUV+ionizing two-segment bookkeeping, the diffuse `ion_bin_of` edge search, and the secondary-ionization band integrals whose $E_0 > 40$ eV branch becomes the COMMON case on a hard field. *Update 2026-10-11:* with stages to VII the band may rise above 100 eV only up to the lowest terminal ionization potential of the elements present (Cl VII 114.2, Ar VII 124.4, Fe VII 125.0 eV); above it the run is refused with the share of the ionizing luminosity beyond that potential.

2. $\bar{g}_{\rm ff}$: raise `NGZ` from 4 to 8 in `cooling_mod` (the van Hoof table already spans the required $\gamma^2$ range; the metal free-free sum then covers the new stages). *Update 2026-10-11:* `NGZ` is now `MAX_ION_STAGES` $-1 = 6$, the highest ion charge carried.

3. `te_max`: allow $10^5$–$10^6$ K for X-ray-heated zones. Any ion used above $10^5$ K needs its Tier-1 fit regenerated over the wider range (a flag in `chianti_cooling.py`, not new code); the bisection itself is range-agnostic.

4. Recommended hard-field defaults, documented in the example inputs: `use_sec_ion=.true.` (a 150 kK field makes $E_0 > 40$ eV photoelectrons routine, and X-ray fields more so) and `launch_sequence='sobol'` (the shielded, partially neutral zones where secondaries act are exactly where launch noise hurts most).

*Gate HS1:* the G0 pattern on the widened band — fixed-state analytic attenuation with a 150 kK blackbody, $\Gamma_{\rm HI/HeI/HeII}$ vs the analytic integrals bin by bin; $Q(>13.6)/Q(>24.6)/Q(>54.4)$ photon-rate ratios against exact Planck integrals to $< 0.1\%$ with aligned edges; and a spot check that the tabulated VFKY96 metal cross sections at 300–500 eV match the published fit values (guards against table-range assumptions in `photo_xsec`).

## 16.4 HS2: registry upward in ion stage

A data operation through the existing fitting code:

Table 12: Target stages and the diagnostics they unlock.

| El. | now→target | new thresholds [eV] | key new lines |
|---|---|---|---|
| C | 4→5 | C IV→V 64.5 | C IV 1548/1550 (resonance — optically thin caveat) |
| N | 3→5 | 47.4, 77.5 | N IV] 1486, N V 1240 |

*Table 12 continued.*

| El. | now→target | new thresholds [eV] | key new lines |
|---|---|---|---|
| O | 3→6 | 54.9, 77.4, 113.9 | [O IV] 25.9 $\mu$m, O IV] 1400, O V] 1218 |
| Ne | 3→6 | 63.4, 97.1, 126.2 | [Ne IV] 2422/2425, [Ne V] 3346/3426 + 14.3/24.3 $\mu$m, [Ne VI] 7.65 $\mu$m |
| S | 4→6 | 47.2, 72.6 | [S IV] 10.5 $\mu$m (new $n$-level file) |
| Ar | 4→6 | 59.8, 75.0 | [Ar V] 7.90/13.1 $\mu$m, [Ar VI] 4.53 $\mu$m |
| Mg | 3→6 | 80.1, 109.3, 141.3 | stage tracking for the PN150 composition (Mg $3 \times 10^{-5}$); [Mg IV] 4.49 $\mu$m, [Mg V] 5.61 $\mu$m as data permit |
| Si | 5→6 | Si V→VI 166.8 | stage tracking for the PN150 composition (Si $3 \times 10^{-5}$) |

Fe/Cl/Ca hold (extend only on diagnostic demand; coronal [Fe VII+] deferred). All targets fit the existing `MAX_ST` = 6 ceiling exactly; `MAXLEV` = 40 / `MAXTR` = 600 and the 600-line `lines_mod` arrays already accommodate the new $n$-level models. Work items, all through `tools/fitting/`:

1. **Element files** (`make_element_data.py`): VFKY96 `PHOTO` rows exist in phfit2 for every target stage; Voronov CI covers all stages; Badnell 2023 RR+DR cover these isoelectronic sequences in the in-tree `clist_K` files (with the CHIANTI `RR2`/`DR2` fallback forms already supported where a fit is absent). **Charge exchange** above stage ∼4 is not in the Kingdon–Ferland fits: adopt the fast Dalgarno-type $\sim 1.92 \times 10^{-9} z$ cm$^3$s$^{-1}$ recombination-only scaling used by Cloudy, with an explicit provenance note in each element file, or document neglect — decide once, apply uniformly (CX matters little in the hot interior where these stages live, but the decision must be recorded).

2. **Tier-1 cooling fits** (`chianti_cooling.py`) for each new stage, fit range matched to the chosen `te_max`.

3. **Tier-2 $n$-level files** for the diagnostic ions ([Ne IV], [Ne V], [O IV], [S IV], [Ar V], [Ar VI], N IV, N V, C IV); CHIANTI v11 carries level and collision data for all of them. Ions without CHIANTI level data are tracked without lines — the established Ar II pattern.

4. **PN abundance example** (`examples/`): the `abunPN150.in` values verbatim, so the PN150 gate input is reproducible from the repository alone.

*Gate HS2:* PyNeb ratio checks ion by ion at the established database-spread tolerances (sub-percent for well-determined optical pairs, up to ∼5–10% for the known compilation-spread classes): the [Ne V] 3426/24.3 $\mu$m temperature pair, the [Ne IV] 2422/2425 and [Ar IV] 4711/4740 density pairs, [O IV] 25.9 $\mu$m against its A-value limit; then an end-to-end hard-field smoke showing every new line in `_lines.txt` with physically sane ratios.
*Update 2026-10-11:* done, wider than the table: every element, Fe, Cl and Ca included, now runs to stage VII (`ELEMENT s 16 7` and likewise), so `MAX_ST` is 7. Charge exchange above $q = 4$ was decided as recorded in item 1: no reaction by default, and the Langevin upper limit (Ferland et al. 1997 form) for $q = 5$, 6 as the option `par%langevin_charge_exchange`.

Items 2 and 3 became one: the atoms of the full CHIANTI 11.0.2 model carry both the line cooling and the lines (Cl VII, Ar II and Ca IV from the CLOUDY Stout files; seven stages without line cooling, listed in the status paragraph at the head of this memo). On a $10^5$ K blackbody with `eion_max` = 150 eV the new transitions receive rates and heating wherever the field reaches them (e.g. O VI→VII at 138 eV). The PyNeb ratio part of the gate is not recorded here.

## 16.5 HS3: He II excited-channel diffuse photons

The PN He III zone is large, and case-B He II recombinations emit ionizing photons beyond the ground continuum already carried (`diffuse_mod` channel 3). The hydrogenic $n = 2$ cascade of He II produces three components, every one of which ionizes H I:

- He II Ly$\alpha$ at 40.8 eV — ionizes H I AND He I, the dominant piece;
- the He II Balmer continuum — its threshold is $Z^2\,\mathrm{Ry}/n^2 = 13.60$ eV, i.e. exactly at the H I edge; the aligned-grid `ion_bin_of` binary search (the 2026-07-18 bug fix) is precisely what makes this land in the correct bin;
- the He II two-photon continuum (0–40.8 eV), of which the $> 13.6$ eV part ionizes — sampled like the He I two-photon branch with its ionizing probability precomputed.

Implementation is the `hei_diffuse` pattern verbatim: a fifth channel in `dif_ch` weighted by the case-B He III recombination rate in each leaf, branch fractions from the hydrogenic 2s/2p cascade shares (Osterbrock & Ferland tables), energies fixed (Ly$\alpha$), threshold-plus-exponential (Balmer continuum), or flat-sampled (two-photon). The quasi-random diffuse launch needs NO new dimensions — the channel pick d7 simply spans five values. Regime note: at HII40 the analogous He I channel moved $T_e$ by only $\sim$1%, but there the He III zone was a small core; in a 150 kK PN the He III region fills much of the nebula and this becomes a first-order energy-budget term.
*Gate HS3:* channel energy conservation (emitted diffuse luminosity equals the case-B $n \geq 2$ recombination power of the He III zone); the PN150 He structure and $T_e$ recorded with the channel off/on, attributed as one lever.
*Update:* done 2026-08-09 as the fifth diffuse channel (`src/heii_cascade_mod.f90`, `par%heii_diffuse`, on by default with the transported continuum when `eion_max` $\geq 40.813$ eV; gate `tests/heii_cascade/`). Two points differ from the draft above: the Balmer (free–bound into $n \geq 2$) continuum is taken from the same free-bound plus free-free tables as the written nebular continuum rather than from a threshold-plus-exponential form, and since 2026-09-26 the $n = 2$ branching comes from the zero-density cascade of the H II/He III case-B coefficients (Section 8.2); the PN150 measurement of the gate is still open.

## 16.6 HS4: PN validation gates

**MOCASSIN PN150/PN75**, with the full-quality protocol that resolved HII40: MOCASSIN re-run CONVERGED (its stored reduced settings — $10^6$ photons, 10 iterations, `convLimit` 0.05, $13^3$ — are a moving target exactly as the HII20/40 baselines were), and MOCHII run with `conv_crit='vol'`, `nnu_ion` 64–128 aligned, $\geq 4 \times 10^7$ packets, `launch_sequence='sobol'`. The PN150 setup is fully specified by the benchmark input quoted in Table 11 (150 kK, $n_H = 3000$ cm$^{-3}$, shell $10^{17}$–$4.07 \times 10^{17}$ cm). Compare, in order of diagnostic power:

1. $T_e$(H II) and the radial $T_e$ profile (PN Te ∼12–18 kK — also exercises the hotter thermal range);
2. the He I/II/III structure — the He II-edge (54.4 eV) photon budget is the PN analog of the He edge discretization that dominated HII40, so the aligned grid must show its value here;
3. the new high-stage fractions O III/IV/V, Ne III/IV/V, N III/IV;
4. the line table: $L$(H$\beta$), He II 4686/H$\beta$, [O III] 5007/4363, [Ne V] 3426, [Ne III] 15.5 $\mu$m, [O IV] 25.9 $\mu$m, [N II] 6584, [S IV] 10.5 $\mu$m, [Ar V].

**Cloudy `pn_paris`** as the 1D cross-check with identical SED, abundances, density, and dust off: zone-by-zone $T_e$ and ion fractions along the radius plus ∼20 integrated ratios. Because Cloudy solves density-dependent level populations in its thermal loop, the PN150 residual BEFORE stage HS7 directly measures how much the low-density Tier-1 fits miss at $n_H$ = 3000 cm$^{-3}$ — that number is the input to HS7, which is why HS4 deliberately precedes it.

*Acceptance:* land inside (or bracket) the published Lexington PN150 cross-code spread, with each residual attributed one lever at a time (binning, diffuse channels, density-dependent cooling, data vintage) exactly as the HII40 campaign did. That spread does not have to be assembled: Jin et al. (2022) Table 4 tabulates 40 PN150 lines for MAPPINGS, CLOUDY, MOCASSIN3D and PH with the median and 1$\sigma$, beside M$^3$ itself at 33$^3$ and 55$^3$ [1], and it covers every line of the list above except [S IV] 10.5 $\mu$m and [Ar V]. Read it with the resolution caveat its own Section 8 attaches: at PN150 the 33$^3$ → 55$^3$ change exceeds 10% on [N I] 5200+5198, [O I] 63.1 $\mu$m, [O I] 6300+6363, [Ne II] 12.8 $\mu$m, Si II] 2335+, [S II] 6716+6731 and [S II] 4069+4076, reaching 40% on [O I] 6300+6363. Two MOCHII-side items are specific to this density. PN150 is the first gate at which the H I two-photon collisional suppression bites — $n_H$ = 3000 cm$^{-3}$ against $n_{crit} \simeq 1.6\times10^4$ cm$^{-3}$ is a 16% reduction of that continuum, where HII20/HII40 at 100 cm$^{-3}$ carry under 1%. And the sulfur lines are on the list, so the $\alpha(S^{2+} \rightarrow S^{+})$ vintage difference of Section 11.3 reaches this gate too and must be separated from geometry before anything is read into [S II] 6716+6731. How much of it reaches here has been measured, and it is not what HII20 would have predicted. **The sulfur offset has been measured at this density, before the gate runs.** CLOUDY's test suite carries the PN150 benchmark as `tsuite/auto/pn_paris.in` — "Paris meeting PN", and its parameters are Jin Table 1 entry for entry (150 kK blackbody, $n_H$ = 3000, $R_{in} = 10^{17}$ cm, He 0.1, C $3.0\times10^{-4}$, N $1.0\times10^{-4}$, O $6.0\times10^{-4}$, Ne $1.5\times10^{-4}$, Mg and Si $3.0\times10^{-5}$, S $1.5\times10^{-5}$, Cl/Ar/Fe off). It was run here unchanged except for lowering `print line faint` (c25.00, 187 zones, 33 s, all 69 monitors within tolerance), and renormalized from its `Ca B` 4861 normalization to H I 4861 (a factor 1.0071). Against the median of Jin Table 4:

| | Med | $\sigma$ | M$^3$ | CLOUDY c25.00 | ratio to median |
|---|---|---|---|---|---|
| [S II] 6716+6731 | 0.359 | 0.050 | 0.311 | 0.6162 | **1.72** |
| [S II] 4069+4076 | 0.064 | 0.012 | 0.049 | 0.1170 | **1.83** |
| [S III] 18.7 $\mu$m | 0.527 | 0.084 | 0.526 | 0.5067 | 0.96 |
| [S III] 33.6 $\mu$m | 0.210 | 0.032 | 0.200 | 0.1879 | 0.89 |
| [S III] 9532+9069 | 2.02 | 0.08 | 2.02 | 1.9853 | 0.98 |

so the offset does not simply carry over from HII20: the [S II] excess is *larger* (1.72 and 1.83 against 1.49 and 1.51) while the [S III] deficit has *gone* (0.96, 0.89, 0.98 against 0.56, 0.56). CLOUDY's own volume-mean ionization says why. At PN150 sulfur is $S^+$ 0.162 / $S^{2+}$ 0.402 / $S^{3+}$ 0.335 / $S^{4+}$ 0.075 / $S^{5+}$ 0.022, so $S^{2+}$ sits in the middle of a chain and

whatever a faster $\alpha(S^{2+} \rightarrow S^{+})$ takes out of it is refilled from $S^{3+}$; at HII20, where the same average is $S^{+}$ 0.61 / $S^{2+}$ 0.39 / $S^{3+}$ 0.000, $S^{2+}$ is the top stage and has nothing above it to draw on, so both members of the pair move. CLOUDY's regression history shows the same asymmetry directly: the 2014-12-14 `update S2+ - S+ DR to Badnell+ 2015 ApJ` moved this model's [S II] 6720 by 0.4412 → 0.499 (+13%) and its [S III] 9531 by only 1.449 → 1.378 (–5%).

*What that means for the gate.* When MOCHII runs PN150 it should be *expected* to land near 1.7–1.8 times the published [S II] with [S III] on the median, and neither is a defect. The corollary matters more: at this density [S III] agreeing with the published spread does *not* clear the sulfur data, because the vintage difference barely shows there. Four other departures of CLOUDY c25.00 from the same median should be subtracted before any MOCHII residual is attributed to geometry or transport: He II 4686 at 0.82× (–6$\sigma$), [Ne III] 3869+3968 at 0.72× (–2.8$\sigma$), $L(\mathrm{H}\beta)$ at 1.08× ($2.897\times10^{35}$ against 2.68), and [O III] 5007+4959 at 1.08×. [O I] 6300+6363 is the one that needs no correction at all: 0.1538 against a median of 0.154.

*And the gate was blocked on HS2, quantitatively.* That same ionization structure puts 10.1% of sulfur above $S^{3+}$ ($S^{4+}$ 0.075 + $S^{5+}$ 0.022 + $S^{6+}$ 0.004), while MOCHII's registry stopped at S IV until 2026-10-11, so $S^{3+}$ was its terminal stage and would have absorbed that 10%. [S IV] 10.5 $\mu$m is meanwhile the strongest sulfur line in the model at 2.46 relative to H$\beta$. The same argument read off the other rows: $Ne^{4+}$ and above, $O^{4+}$ and above, and the Mg/Si stage tracking are all populated here. Since 2026-10-11 every element runs to stage VII (`data/atomic/element_s.txt`, `ELEMENT s 16 7`), so all of Cloudy's sulfur stages above are carried, $S^{6+}$ as the terminal one; the PN150 run is among the pending `m13` runs.

## 16.7 HS5: AGN SED input

An `ion_spectrum='agn'` preset implementing the Cloudy continuum (`parse_agn.cpp`; Mathews–Ferland form):

$$f_\nu \;=\; \nu^{\alpha_{\mathrm{uv}}}\, e^{-h\nu/kT_{\mathrm{bump}}}\, e^{-kT_{\mathrm{IR}}/h\nu} \;+\; a\,\nu^{\alpha_{\mathrm{x}}}, \tag{8}$$

with the X-ray normalization $a$ fixed by $\alpha_{\mathrm{ox}}$ through the 2500 Å–2 keV energy ratio (the 403.3 factor in the Cloudy source) and the standard defaults $\alpha_{\mathrm{uv}} = -0.5$, $\alpha_{\mathrm{x}} = -1$, IR cutoff at $kT_{\mathrm{IR}} = 0.01$ Ryd. It enters `ion_band_mod` exactly like the Planck function — a spectral density integrated on the 32-point sub-grid in each band bin — and is normalized by the existing derive-or-rescale luminosity rules. The tabulated-file route (`spectrum_type='per_ev'` etc.) remains the reference path and already reproduces the `agn_lex00` gate SEDs verbatim (Table 11); the preset adds convenience and reproducibility for parameter studies. Also add an ionization-parameter helper to the rank-0 log, $U = Q_{\mathrm{ion}}/(4\pi r^2 n_{\mathrm{H}} c)$ evaluated at a reference radius, so slab setups map onto Cloudy inputs without hand conversion.

*Gate HS5:* the bin-integrated preset spectrum against Cloudy `save continuum` output for identical parameters (shape agreement to ∼1% per bin); $Q$-ratio integrals exact; the derive/rescale bookkeeping re-verified on the preset (the ISRF-preset test pattern).

## 16.8 HS6: inner-shell absorption and Auger (AGN track)

The one genuine framework extension of the plan, in four parts:

1. **Shell-resolved opacity.** Above each K/L edge the total cross section becomes the VFKY96 outer-shell fit plus the Verner & Yakovlev (1995) inner-shell partials, $\sigma_{\mathrm{tot}}(E) = \sigma_{\mathrm{outer}}(E) + \sum_s \sigma_s(E)$. The `sigma_vy95` evaluator and the

`PHOTO2` element-file rows already exist (the Cl path); the extension is carrying several shells with their edges in the aligned-band threshold list. *Update 2026-09-24, restored 2026-10-11:* this part is done, in the form `phfit2` itself uses, which is not the sum written above: below the first inner-shell edge $E_{\rm inn}$ the outer-shell fit alone, at and above it the sum of the Verner & Yakovlev subshell fits of *all* occupied subshells, the outer one included (element-file rows `PHOTO`, `PHINNER`, `PHSHELL`; `PHOTO2` is gone), for the opacity, the rates and the shell-resolved heating. The tree of October 9–11 had replaced it with the outer shell alone, which deleted valid opacity; 2026-10-11 restored the sum, and it is now the only form (`par%metal_inner_shell_absorption` is refused; gate `tests/photo_shells`, worst $2.3 \times 10^{-6}$). The product of every absorption is the next stage. Auger yields, the stage matrix and the Auger-electron energetics (parts 2–4) remain open, planned as `par%auger_ionization`, and they already matter inside the default band: the Mg I $2p$, Ca I $3p$ and Fe I–III $3p$ vacancies emit a second electron with probability $\geq 0.995$ (Kaastra & Mewe 1993 as tabulated by CLOUDY), which the one-stage bookkeeping does not carry (a known defect, marked at `species_cross_section`).

2. **Auger yields.** Kaastra & Mewe (1993) electrons per vacancy, shell-resolved, read from the Cloudy data file format (`mewe_nelectron.dat`: element, ion, shell, yield distribution): an inner-shell ionization of stage $i$ lands at stage $i+n_{\rm e}$(shell) with the tabulated probability distribution over $n_{\rm e}$. Fluorescence (K$\alpha$ photon emission instead of one Auger electron, `mewe_fluor.dat`) is a small branch for low-$Z$ elements; the first increment deposits it as local heat with a documented note rather than transporting the line photon.

3. **Cascade generalization.** `species_fractions` currently solves the strict adjacent-stage chain as a product form. Auger couples stage $i$ to $i+2, i+3, \ldots$, so each element's stationary balance becomes a small $n_{\rm stage} \times n_{\rm stage}$ matrix problem ($M\mathbf{x} = 0$ with $\sum_j x_j = 1$); the partial-pivot elimination already used by `nlevel_mod` is reused directly. The chain path is KEPT as the default and the matrix path activates only when an inner-shell channel is present — the off-path stays arithmetically identical, the house rule for every extension.

4. **Energetics.** The Auger electron carries $E_0 = h\nu - E_{\rm th,shell} - \sum E_{\rm secondary\ vacancies}$ and is fast; it enters the EXISTING Shull–van Steenberg partition through the absorber-resolved excess-energy integrals with no new heating physics. Compton remains excluded at $\leq$500 eV.

*Gate HS6:* the X-ray-illuminated slab against the Cloudy `agn_lex00_u{m1,0,1}` set (three ionization parameters at $n_{\rm H} = 10^5$ cm$^{-3}$, tabulated SED, stop column $10^{16}$ cm$^{-2}$): ion-fraction profiles of C/N/O/Ne through the partially ionized zone, the $T_{\rm e}$ profile, and the H/He recombination lines, within the published Lexington-2000 cross-code spread. A secondary check: with Auger artificially disabled the slab must reproduce the HS2-era solution (isolates the new coupling).

## 16.9 HS7: density-dependent metal cooling in the thermal loop

Required at PN densities. The coolants that dominate at nebular temperatures quench in a specific order because their critical densities span five decades (approximate values at $10^4$ K): [C II] 158 $\mu$m $n_{\rm crit,e} \sim 50$, [N II] 122 $\mu$m $\sim 3 \times 10^2$, [O III] 88 $\mu$m $\sim 5 \times 10^2$, [O III] 52 $\mu$m $\sim 3.5 \times 10^3$, [O II] 3729 $\sim 3 \times 10^3$, [S II] 6716 $\sim 2 \times 10^3$, [N II] 6584 $\sim 9 \times 10^4$, [O III] 5007 $\sim 7 \times 10^5$. At the PN150 density ($n_{\rm H} = 3000$, $n_{\rm e}$ up to $\sim$7000 cm$^{-3}$ in the He III zone) the IR fine-structure

lines are already partly quenched while the optical forbidden lines are not — the low-density Tier-1 fits therefore OVERESTIMATE cooling and bias $T_e$ low, with the error growing toward compact PN and NLR densities.
Primary design, keeping "fitted offline, evaluated online":

1. `chianti_cooling.py` already solves the full $n$-level system to build its fits; extend it to emit $\Lambda(T, n_e)$ tables on a log-$n_e$ grid ($\sim$7 nodes, $10^1$–$10^7$ cm$^{-3}$) $\times$ the existing log-$T$ range, one block appended to each `cooling_tier1_*.txt` behind a format-version flag (old files keep working).

2. `species_mod::metal_cooling` evaluates bilinear-in-$(\log T, \log n_e)$; the memory cost is trivial ($\sim 60\times 7$ values for each ion) and the evaluation stays a handful of multiply-adds inside the bisection — the `species_ne` cache pattern applies unchanged.

3. The lowest-density node MUST reproduce the current fits to fit accuracy (regression guard), so HII20/HII40 and every recorded gate move by $< 0.1\%$.

4. Verification: an in-loop $n$-level solve for the dominant coolants on a small model, confirming table interpolation error $< 1\%$ in $\Lambda$ over the used $(T, n_e)$ range; the H-impact PDR cooling terms (`metal_cooling_H`) stay as they are (their two-level low-density form is correct in the PDR regime they serve). *Update 2026-09-24:* that last judgment was wrong and is superseded. The two-level form carried an unsourced [O I] coefficient 0.20–0.46 of Abrahamsson et al. (2007), no C I channel, and a saturation denominator separate from the electrons'; `metal_cooling_H` now takes the increment of one $n$-level solve with $R = n_e q^e + n_{HI} q^H + A$.

*Gate HS7:* a two-zone analytic check (one low- and one high-density zone against the exact $n$-level cooling); PN150 $T_e$ with the tables off/on (the residual measured at HS4 must shrink accordingly); HII20/HII40 unchanged to $< 0.1\%$.
*Update:* done 2026-07-26, ahead of HS2–HS4 and in a different form: `par%cooling_model = 'local_ne'` (the default) builds the $\Lambda(T_e, n_e)$ suppression table at setup from the $n$-level atoms (`src/nlevel_cooling_mod.f90`, $\log_{10} T_e = 2$–$6$ and $\log_{10} n_e = -6$–$10$) instead of appending it to the cooling files, and the zero-density fits remain as `'low_density'`. The expectation of item 3 did not hold: the Lexington benchmarks at $n_H = 100$ moved by +316 K (HII40, 7894 to 8210 K) and +648 K (HII20, 6376 to 7024 K), because [C II] 158 $\mu$m and the [O III] far-infrared lines are already partly quenched there (`docs/MoCHII_cooling_analysis.pdf`). The PN150 part of the gate is open, and the HS4 residual can no longer be measured before HS7.

## 16.10 Deferred items

Recorded once so they are conscious exclusions, not oversights: Compton heating/cooling and Thomson-scattering transport ($\gtrsim$0.5 keV; Cloudy is the reference implementation); Fe K fluorescence photon transport (the yield enters HS6 as local heat); coronal ions beyond $\sim$stage 8 ([Fe VII] 6087, [Fe X] 6374 need Fe stages 8–11 and a hotter thermal range); X-ray and UV resonance-line transfer (LaRT territory); grain destruction/sputtering in hard radiation fields (the PAH survival switches are a partial hook, not a destruction model); and time dependence / finite source age, where MAPPINGS remains the 1D reference.

### 16.11 Suggested order

| Order | Stage | Framework risk | Blocking |
|---|---|---|---|
| 1 | HS1 band headroom | low | gates HS2+ |
| 2 | HS2 registry upward | none (data) | gates HS4 |
| 3 | HS3 He II diffuse channels | low | sharpens HS4 |
| 4 | HS4 PN gates (MOCASSIN + Cloudy) | none | PN track done |
| 5 | HS7 density-dependent cooling | medium | PN accuracy |
| 6 | HS5 AGN SED preset | low | gates HS6 |
| 7 | HS6 inner-shell/Auger | high (cascade) | AGN track done |

HS7 can run in parallel with HS2–HS4 (independent code paths); the first PN150 pass deliberately precedes HS7, so it measures how much of the PN residual the density-dependent cooling must explain — the same one-lever-at-a-time attribution that worked for HII40. (As drafted; HS7 and HS3 have since been done, so that PN150 pass now runs with `par%cooling_model = 'low_density'` if the attribution is still wanted.) As of 2026-10-11, HS2, HS3, HS7 and HS6 part 1 are done, the MOCHII PN150 run of HS4 is running (`m13`), and HS1 beyond the terminal-band refusal, HS5 and HS6 parts 2–4 are open.

## 17. Feasibility: time dependence, cooling flows, and shocks in MOCHII

Section 17. addresses the three capabilities where the public M$^3$ 3D driver and MOCHII both trail MAPPINGS V 5.2.1: the connected time-dependent ionization history of P6/P7, the 1D cooling flow, and the S5 steady radiative shock with an iterated precursor (§3.2). The question is not whether these are useful — they plainly are — but how much each would cost to add to MOCHII, given its architecture. The three answers differ sharply: time dependence is a bounded project that the design accommodates naturally, a cooling flow rides on top of it, and a radiative shock is a separate one-dimensional radiation-hydrodynamic calculation that the Monte Carlo transport engine does little to help.

### 17.1 The architectural pivot

MOCHII's iteration loop transports Monte Carlo packets, tallies the cell radiation field, solves the *equilibrium* ionization and thermal balance in each cell (`solve_ion_cell`: a damped fixed point on $n_e$ with a bisection on $T_e$), refreshes the opacity, and repeats until converged. The state solver is therefore a *root finder*: given the rates it returns the steady state. It integrates no $\mathrm{d}n_i/\mathrm{d}t$, carries no velocity field, and solves no fluid equations. That single fact sets the cost of each addition: one reuses the root finder almost unchanged, one adds a time integrator beside it, and one needs what the code does not have at all.

### 17.2 Time-dependent ionization: moderate, and the natural fit

Almost everything a finite-lifetime calculation needs already exists. The transport, the photoionization, recombination, collisional-ionization, and charge-exchange rate coefficients (already evaluated cell by cell), the tally, the octree, and the

diffuse field are all reusable without change. The cell solver is already "given the rates, find the state"; the new piece is a sibling routine, "given the rates, the old state, and $\Delta t$, find the new state."

What is genuinely new is threefold. First, a stiff ordinary-differential integrator for the ionic populations and, where the thermal time is not short compared with the ionization time, the electron temperature — a backward-Euler or semi-implicit step in each cell, of the kind MAPPINGS V 5.2.1's `timion` performs, rather than the present root find. Second, an outer time loop with a step controller keyed to the atomic, cooling, and photon-absorption times. Third, a source light curve or a finite on/off lifetime as input.

The design accommodates this cleanly through operator splitting. MOCHII already performs one transport pass for each iteration, so the natural structure is an outer radiation step that holds $J_\nu$ fixed while the chemistry is substepped in each cell, with the radiation field recomputed once the ionization has advanced enough to change the opacity. An ionization front that advances while a source brightens, or recedes after it switches off, is a reachable target. The hard part is the stiff integrator and the step control, not the infrastructure.

## 17.3 Cooling flows: depends on which cooling flow

The term carries two meanings, and they price very differently.

In the MAPPINGS V 5.2.1 sense — a parcel of gas that cools radiatively as it moves down a prescribed density and velocity history — a cooling flow is a special case of time-dependent cooling on a fixed trajectory: switch off (or freeze) the photoionizing field, and integrate the non-equilibrium ionization and the radiative losses down a temperature ramp. Once the time-dependent solver of the previous subsection exists, this is a modest addition, because the cooling terms it needs — recombination, free–free, and the metal-line coolants through the Tier-1 fits and the Tier-2 $n$-level solve — are already present. Its one real prerequisite was the density-dependent cooling flagged as HS7 and in §8.5: a cooling flow crosses coolant critical densities, so a $\Lambda(T_\mathrm{e}, n_\mathrm{e})$ that suppresses forbidden lines correctly is required for it to be quantitative; that table exists since 2026-07-26 (`'local_ne'`).

In the cluster or galaxy sense — a self-consistent inflow driven by its own radiative losses, with gravity and hydrodynamics — a cooling flow is a fluid-dynamics problem and lies outside the present scope without a hydrodynamic solver. The comparison with MAPPINGS V 5.2.1 concerns the first meaning.

## 17.4 Radiative shocks: hard, and off the architecture

The MAPPINGS V 5.2.1 S5 shock solves the Rankine–Hugoniot jump conditions and then integrates the post-shock cooling zone as a one-dimensional flow — mass, momentum, and energy conservation along the flow coordinate with radiative losses — while iterating an upstream photoionized precursor computed from the shock's own radiation to self-consistency. Its essence is a one-dimensional radiation-hydrodynamic flow, not a three-dimensional transport problem.

This is the poor fit. MOCHII has no velocity field, no fluid equations, and no flow coordinate. The post-shock structure is gas that moves and compresses; the static octree grid and the Monte Carlo photon transport represent none of that, and three-dimensional Monte Carlo transport buys almost nothing for a one-dimensional steady shock. The honest route is a separate one-dimensional radiative-shock module, a companion solver in the spirit of the 1D references MOCHII

already uses for its gates: it would reuse the atomic and cooling data but not the octree or the Monte Carlo engine, and it would carry the jump conditions, an adaptive cooling-zone integrator, and the precursor iteration itself. That is real work, largely decoupled from the transport code, and — this is the key qualification — it would be a 1D shock, not a 3D radiative shock. A genuine three-dimensional radiative shock needs actual radiation hydrodynamics, which is a different code, not an extension of this one.

### 17.5 Summary and suggested order

| Capability | Difficulty | Why |
|---|---|---|
| Time-dependent ionization | Moderate | Reuses all transport and rates; new work is a stiff cell ODE, a time loop, and a light curve. Operator splitting maps onto the existing one-transport-for-each-iteration loop. |
| Cooling flow (MAPPINGS V 5.2.1 sense) | Moderate | A prescribed-trajectory special case of the time-dependent solver; uses the density-dependent $\Lambda(T_e, n_e)$ of HS7 (in place). |
| Cooling flow (cluster sense) | Out of scope | Requires gravity and hydrodynamics. |
| Radiative shock (S5) | Hard | A separate 1D radiation-hydrodynamic module; the octree and Monte Carlo engine do not apply. Not a 3D shock. |

A sensible order is: (1) time-dependent ionization first, as the most natural addition and the foundation for the rest; (2) the density-dependent cooling of HS7, needed for both PN accuracy and cooling flows (done, 2026-07-26); (3) the MAPPINGS V 5.2.1-sense cooling flow as a thin layer on top of (1) and (2); and (4) a decoupled 1D radiative-shock module last, and only if a shock diagnostic is actually required. The first three share the same time-integration and cooling infrastructure and reinforce the PN and hard-spectrum work of §16.; the fourth stands apart.

## 18. Implementation evidence consulted

The principal local sources used for this audit were:

- MOCHII: `md/PLAN.md`, `docs/MoCHII_physics.tex`, `docs/MoCHII_UserGuide.tex`, the transport/grid modules under `src/`, and the quantitative gates under `tests/`.
- M$^3$: `README.md`, `lab/data/ATDAT.txt`, `src/mastercode/montph7.f`, `montph7_dust.f`, `crosssections.f`, `mcteequi.f`, `teequi2.f`, `equion.f`, `cool.f`, `multilevel.f`, `grainpar.f`, `hgrains.f`, `hpahs.f`, `dusttemp.f`, `timion.f`, and `shock2--shock5.f`.
- MAPPINGS V 5.2.1: `README.md`, `src/mappings.f`, `photo6.f`, `photo7.f`, `teequi.f`, `teequi2.f`, `timion.f`, `timtqui.f`, `cool.f`, `dusttemp.f`, `shock5.f`, `neqc.f`, `coolc.f`, `transferline.f`, `lab/mapStd.prefs`, `lab/mapFull.prefs`, and `docs/DocumentionOverview.html`.

The public M$^3$ source contains several historical or experimental driver variants. Statements in this memo refer to the main `mastercode/montph7.f` and explicitly selected `montph7_dust.f` paths unless otherwise stated. Statements about MAPPINGS V 5.2.1 refer to the connected public 5.2.1 `map52` P6/P7/S5/CC/NC/PP paths, not merely to routines present in the directory.

# References

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[3] Sutherland, R. S., & Dopita, M. A. 1993, "Cooling Functions for Low-density Astrophysical Plasmas," ApJS, 88, 253.

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[7] Badnell, N. R. 2006, "Radiative Recombination Data for Modeling Galactic Formation, Evolution, and Star Formation," ApJS, 167, 334.

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[9] Verner, D. A., Ferland, G. J., Korista, K. T., & Yakovlev, D. G. 1996, "Atomic Data for Astrophysics. II. New Analytic Fits for Photoionization Cross Sections of Atoms and Ions," ApJ, 465, 487.